Final_Project_Notebook

June 11, 2022

1 Imports

[1]:
```
import pandas as pd
import datetime
import re
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from xgboost import XGBClassifier
import numpy as np
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import mean_squared_error
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn import tree
from sklearn.ensemble import RandomForestClassifier
from tqdm import tqdm
from sklearn import metrics
from pygam import LogisticGAM
from sklearn import tree
from statsmodels.stats.outliers_influence import variance_inflation_factor
from sklearn.svm import SVC
from sklearn import naive_bayes
from sklearn.preprocessing import MinMaxScaler
from xgboost import XGBRFClassifier
from sklearn.model_selection import GridSearchCV
```

```
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/compat.py:36:
FutureWarning: pandas.Int64Index is deprecated and will be removed from pandas
in a future version. Use pandas.Index with the appropriate dtype instead.
  from pandas import MultiIndex, Int64Index
```

2 Loading the Data

[2]:
```
historicalWeather = pd.read_csv('HistoricalWeather.csv')
historicalWeatherForecasts = pd.read_csv('HistoricalWeatherForecasts.csv')
historicalWildfires = pd.read_csv('Historical_Wildfires.csv')
landClass = pd.read_csv('LandClass.csv')
vegetationIndex = pd.read_csv('VegetationIndex.csv')
```

3 Creating the Data Set

[3]:
```
historicalWeather_pivot = historicalWeather.pivot(index=['Date']+['Region'],␣
↪columns=['Parameter'])
historicalWeather_pivot = historicalWeather_pivot.dropna()
historicalWeather_pivot.head()
historicalWeather_pivot = historicalWeather_pivot.drop(columns=[('count()[unit:␣
↪km^2]','Precipitation'),
                                ('count()[unit: km^2]', 'RelativeHumidity'),
                                 ('count()[unit: km^2]', 'SoilWaterContent'),
                                 ('count()[unit: km^2]',   'SolarRadiation'),
                                 ('count()[unit: km^2]',      'Temperature')])
```

[4]:
```
reform_dates = []
for i in historicalWildfires.iterrows():

    region = i[1][0]
    date = i[1][1]
    month = re.search('^(\d+)/', date)
    month = month.group(1)
    day = re.search('/(\d+)/', date)
    day = day.group(1)
    year = re.search('/(\d+)$', date)
    year = year.group(1)

    date = datetime.datetime(int(year), int(month), int(day))

    reform_dates.append([date.strftime('%Y-%m-%d'), region])

historicalWeather_pivot['fire01'] = [1 if [i[0][0], i[0][1]] in reform_dates␣
↪else 0 for i in tqdm(historicalWeather_pivot.iterrows())]
```

```
40404it [00:20, 1957.92it/s]
```

[5]:
```
historicalWildfires.head()
```

[5]:
```
   Region      Date  Estimated_fire_area  Mean_estimated_fire_brightness  \
0     NSW  1/4/2005              8.68000                      312.266667
```

```
1    NSW  1/5/2005            16.61125                      322.475000
2    NSW  1/6/2005             5.52000                      325.266667
3    NSW  1/7/2005             6.26400                      313.870000
4    NSW  1/8/2005             5.40000                      337.383333

   Mean_estimated_fire_radiative_power  Mean_confidence  Std_confidence  \
0                            42.400000        78.666667        2.886751
1                            62.362500        85.500000        8.088793
2                            38.400000        78.333333        3.214550
3                            33.800000        92.200000        7.529940
4                           122.533333        91.000000        7.937254

   Var_confidence  Count Replaced
0        8.333333      3        R
1       65.428571      8        R
2       10.333333      3        R
3       56.700000      5        R
4       63.000000      3        R
```

[6]:
```
name_of_index = []
for i in historicalWeather_pivot.index:
    if i[1] == 'NSW':
        name_of_index.append(0)
    if i[1] == 'NT':
        name_of_index.append(1)
    if i[1] == 'QL':
        name_of_index.append(2)
    if i[1] == 'SA':
        name_of_index.append(3)
    if i[1] == 'TA':
        name_of_index.append(4)
    if i[1] == 'VI':
        name_of_index.append(5)
    if i[1] == 'WA':
        name_of_index.append(6)

historicalWeather_pivot.insert(0, 'NameToNum', name_of_index)
name_dummies = pd.get_dummies(historicalWeather_pivot.NameToNum)
historicalWeather_pivot = pd.concat([name_dummies, historicalWeather_pivot],␣
 ↪axis=1)
historicalWeather_pivot
```

[6]:
```
                    0  1  2  3  4  5  6  (NameToNum, )  \
Date       Region
2005-01-01 NSW     1  0  0  0  0  0  0              0
           NT      0  1  0  0  0  0  0              1
           QL      0  0  1  0  0  0  0              2
```

```
           SA      0  0  0  1  0  0  0              3
           TA      0  0  0  0  1  0  0              4
…                .. .. .. .. .. .. ..            …
2020-10-31 QL      0  0  1  0  0  0  0              2
           SA      0  0  0  1  0  0  0              3
           TA      0  0  0  0  1  0  0              4
           VI      0  0  0  0  0  1  0              5
           WA      0  0  0  0  0  0  1              6

                   (count()[unit: km^2], WindSpeed)  (min(), Precipitation)  \
Date       Region
2005-01-01 NSW                         8.002343e+05                0.000000
           NT                          1.357561e+06                0.000000
           QL                          1.730605e+06                0.000000
           SA                          9.913151e+05                0.000000
           TA                          6.778526e+04                0.003973
…                                            …                        …
2020-10-31 QL                          1.730605e+06                0.000000
           SA                          9.913151e+05                0.000000
           TA                          6.778526e+04                0.000703
           VI                          2.294532e+05                0.000000
           WA                          2.528546e+06                0.000000

                   …  (mean(), SolarRadiation)  (mean(), Temperature)  \
Date       Region  …
2005-01-01 NSW     …                 26.749389              27.341182
           NT      …                 19.781791              29.881492
           QL      …                 27.056979              28.842866
           SA      …                 27.142643              30.793675
           TA      …                 26.755711              11.788805
…                 …                     …                      …
2020-10-31 QL      …                 28.745128              24.777835
           SA      …                 26.530568              18.947783
           TA      …                 21.782732              11.648813
           VI      …                 19.553751              13.167147
           WA      …                 26.746413              28.090392

                   (mean(), WindSpeed)  (variance(), Precipitation)  \
Date       Region
2005-01-01 NSW                 3.323550                     0.028362
           NT                  5.296892                   546.059262
           QL                  3.483753                    35.641257
           SA                  4.657538                     0.042837
           TA                  5.408138                    12.068597
…                                 …                           …
2020-10-31 QL                  3.137745                     4.229566
           SA                  3.548225                     0.000492
```

```
           TA                 2.501697                     0.243603
           VI                 3.838360                     3.148454
           WA                 4.373947                     2.097161

                   (variance(), RelativeHumidity)  \
Date       Region
2005-01-01 NSW                        253.559937
           NT                         584.201131
           QL                         403.134377
           SA                         246.044713
           TA                         111.754034
…                                          …
2020-10-31 QL                         316.368911
           SA                         316.308826
           TA                          10.936866
           VI                          44.766480
           WA                         309.018764

                   (variance(), SoilWaterContent)  \
Date       Region
2005-01-01 NSW                          0.007758
           NT                           0.026743
           QL                           0.012679
           SA                           0.001917
           TA                           0.007121
…                                          …
2020-10-31 QL                           0.006935
           SA                           0.005425
           TA                           0.009317
           VI                           0.005050
           WA                           0.001419

                   (variance(), SolarRadiation)  (variance(), Temperature)  \
Date       Region
2005-01-01 NSW                         6.078587                  18.562212
           NT                         58.942658                  12.920252
           QL                         29.500832                  13.792599
           SA                          7.914246                  34.799336
           TA                         12.826400                   4.912013
…                                          …                          …
2020-10-31 QL                          4.796284                   8.811943
           SA                         17.093209                   9.466862
           TA                          8.718983                   2.388990
           VI                          9.917196                   4.088503
           WA                          5.986063                  15.293953

                   (variance(), WindSpeed)  (fire01, )
```

```
Date       Region
2005-01-01 NSW                    0.850048           0
           NT                     1.930014           1
           QL                     0.883048           1
           SA                     1.655908           0
           TA                     2.963118           0
…                                    …          …
2020-10-31 QL                     0.549077           1
           SA                     0.640926           0
           TA                     0.859296           0
           VI                     1.019079           0
           WA                     1.873607           1

[40404 rows x 34 columns]
```

[7]:
```
#rename columns to remove problem characters
historicalWeather_pivot.columns = ([('NSW'),
                                    ('NT'),
                                    ('QL'),
                                    ('SA'),
                                    ('TA'),
                                    ('VI'),
                                    ('WA'),
                                    ('NameToNum'),
                                    ('count',        'Area'),
                                    ('min',    'Precipitation'),
                                    ('min', 'RelativeHumidity'),
                                    (              'min', 'SoilWaterContent'),
                                    (              'min',   'SolarRadiation'),
                                    (              'min',      'Temperature'),
                                    (              'min',        'WindSpeed'),
                                    (              'max',    'Precipitation'),
                                    (              'max', 'RelativeHumidity'),
                                    (              'max', 'SoilWaterContent'),
                                    (              'max',   'SolarRadiation'),
                                    (              'max',      'Temperature'),
                                    (              'max',        'WindSpeed'),
                                    (             'mean',    'Precipitation'),
                                    (             'mean', 'RelativeHumidity'),
                                    (             'mean', 'SoilWaterContent'),
                                    (             'mean',   'SolarRadiation'),
                                    (             'mean',      'Temperature'),
                                    (             'mean',        'WindSpeed'),
                                    (         'variance',    'Precipitation'),
                                    (         'variance', 'RelativeHumidity'),
                                    (         'variance', 'SoilWaterContent'),
                                    (         'variance',   'SolarRadiation'),
```

```
                                    (          'variance',       'Temperature'),
                                    (          'variance',         'WindSpeed'),
                                    ('fire01')])
historicalWeather_pivot = historicalWeather_pivot.drop(columns='NameToNum')
historicalWeather_pivot
```

[7]:

```
                    NSW  NT  QL  SA  TA  VI  WA  (count, Area)  \
Date       Region
2005-01-01 NSW       1   0   0   0   0   0   0   8.002343e+05
           NT        0   1   0   0   0   0   0   1.357561e+06
           QL        0   0   1   0   0   0   0   1.730605e+06
           SA        0   0   0   1   0   0   0   9.913151e+05
           TA        0   0   0   0   1   0   0   6.778526e+04
…                 …  ..  ..  ..  ..  ..  ..            …
2020-10-31 QL        0   0   1   0   0   0   0   1.730605e+06
           SA        0   0   0   1   0   0   0   9.913151e+05
           TA        0   0   0   0   1   0   0   6.778526e+04
           VI        0   0   0   0   0   1   0   2.294532e+05
           WA        0   0   0   0   0   0   1   2.528546e+06

                    (min, Precipitation)  (min, RelativeHumidity)  …  \
Date       Region                                                  …
2005-01-01 NSW                  0.000000                 13.877194  …
           NT                   0.000000                 14.558820  …
           QL                   0.000000                 14.443199  …
           SA                   0.000000                 10.618136  …
           TA                   0.003973                 43.906574  …
…                                  …                        …  …
2020-10-31 QL                   0.000000                 12.749949  …
           SA                   0.000000                 12.512371  …
           TA                   0.000703                 69.801094  …
           VI                   0.000000                 66.459290  …
           WA                   0.000000                  9.272744  …

                    (mean, SolarRadiation)  (mean, Temperature)  \
Date       Region
2005-01-01 NSW                   26.749389            27.341182
           NT                    19.781791            29.881492
           QL                    27.056979            28.842866
           SA                    27.142643            30.793675
           TA                    26.755711            11.788805
…                                    …                    …
2020-10-31 QL                    28.745128            24.777835
           SA                    26.530568            18.947783
           TA                    21.782732            11.648813
           VI                    19.553751            13.167147
           WA                    26.746413            28.090392
```

```
                    (mean, WindSpeed)  (variance, Precipitation)  \
Date       Region
2005-01-01 NSW              3.323550                   0.028362
           NT               5.296892                 546.059262
           QL               3.483753                  35.641257
           SA               4.657538                   0.042837
           TA               5.408138                  12.068597
…                                …                         …
2020-10-31 QL               3.137745                   4.229566
           SA               3.548225                   0.000492
           TA               2.501697                   0.243603
           VI               3.838360                   3.148454
           WA               4.373947                   2.097161

                    (variance, RelativeHumidity)  (variance, SoilWaterContent)  \
Date       Region
2005-01-01 NSW                       253.559937                       0.007758
           NT                        584.201131                       0.026743
           QL                        403.134377                       0.012679
           SA                        246.044713                       0.001917
           TA                        111.754034                       0.007121
…                                         …                              …
2020-10-31 QL                        316.368911                       0.006935
           SA                        316.308826                       0.005425
           TA                         10.936866                       0.009317
           VI                         44.766480                       0.005050
           WA                        309.018764                       0.001419

                    (variance, SolarRadiation)  (variance, Temperature)  \
Date       Region
2005-01-01 NSW                       6.078587                18.562212
           NT                       58.942658                12.920252
           QL                       29.500832                13.792599
           SA                        7.914246                34.799336
           TA                       12.826400                 4.912013
…                                       …                        …
2020-10-31 QL                        4.796284                 8.811943
           SA                       17.093209                 9.466862
           TA                        8.718983                 2.388990
           VI                        9.917196                 4.088503
           WA                        5.986063                15.293953

                    (variance, WindSpeed)  fire01
Date       Region
2005-01-01 NSW                   0.850048       0
           NT                    1.930014       1
```

```
            QL                    0.883048       1
            SA                    1.655908       0
            TA                    2.963118       0
…                                    …      …
2020-10-31  QL                    0.549077       1
            SA                    0.640926       0
            TA                    0.859296       0
            VI                    1.019079       0
            WA                    1.873607       1

[40404 rows x 33 columns]
```

4 VIF Analysis

[9]:
```
X = historicalWeather_pivot[list(set(historicalWeather_pivot.columns) -␣
↪set('fire01'))]
vif_ = pd.DataFrame()
vif_["feature"] = X.columns

vif_["VIF"] = [variance_inflation_factor(X.values, i) for i in range(len(X.
↪columns))]

print(vif_)
```

```
/home/charlie/anaconda3/lib/python3.9/site-packages/pandas/core/common.py:241:
VisibleDeprecationWarning: Creating an ndarray from ragged nested sequences
(which is a list-or-tuple of lists-or-tuples-or ndarrays with different lengths
or shapes) is deprecated. If you meant to do this, you must specify
'dtype=object' when creating the ndarray.
  result = np.asarray(values, dtype=dtype)
/home/charlie/anaconda3/lib/python3.9/site-
packages/statsmodels/stats/outliers_influence.py:195: RuntimeWarning: divide by
zero encountered in double_scalars
  vif = 1. / (1. - r_squared_i)
                         feature           VIF
0               (min, WindSpeed)  4.977948e+00
1                            NSW           inf
2            (mean, Temperature)  4.457037e+01
3        (max, RelativeHumidity)  2.960226e+00
4              (mean, WindSpeed)  7.356593e+00
5        (min, SoilWaterContent)  2.623690e+00
6          (min, SolarRadiation)  7.710638e+00
7                             WA           inf
8                         fire01  1.921411e+00
9        (min, RelativeHumidity)  2.106018e+01
10                            QL           inf
```

```
11     (variance, Precipitation)  3.603813e+00
12      (mean, SoilWaterContent)  1.571319e+01
13        (mean, SolarRadiation)  3.219168e+01
14      (mean, RelativeHumidity)  1.946206e+01
15                 (count, Area)           inf
16       (variance, Temperature)  5.731700e+00
17          (max, Precipitation)  4.052888e+00
18                            VI           inf
19         (variance, WindSpeed)  3.633465e+00
20                            SA           inf
21                            TA  9.007199e+15
22                            NT           inf
23          (min, Precipitation)  1.611378e+00
24            (max, Temperature)  4.754164e+01
25  (variance, SoilWaterContent)  5.015106e+00
26    (variance, SolarRadiation)  4.974307e+00
27       (max, SoilWaterContent)  3.457697e+00
28              (max, WindSpeed)  2.886077e+00
29         (mean, Precipitation)  4.550328e+00
30  (variance, RelativeHumidity)  5.536556e+00
31            (min, Temperature)  2.192228e+01
32         (max, SolarRadiation)  2.885985e+01
```

Running the following cells made each of the models weaker and therefore this section was NOT run:

[13]:
```
# count = 0
# for i in vif_['VIF']:
#     if i > 10:
#         historicalWeather_pivot = historicalWeather_pivot.
 ↪drop(columns=[list(vif_['feature'])[count]])
#     count += 1

# print(vif_)
```

```
                        feature        VIF
0        (max, RelativeHumidity)   2.960226
1          (mean, Precipitation)   4.550328
2     (variance, SolarRadiation)   4.974307
3                            NT        inf
4          (min, SolarRadiation)   7.710638
5       (mean, RelativeHumidity)  19.462064
6   (variance, SoilWaterContent)   5.015106
7          (max, SolarRadiation)  28.859853
8          (variance, WindSpeed)   3.633465
9        (min, SoilWaterContent)   2.623690
10              (min, WindSpeed)   4.977948
11                        fire01   1.921411
```

```
12        (mean, SolarRadiation)  32.191678
13       (max, SoilWaterContent)   3.457697
14                            WA        inf
15                            TA        inf
16       (variance, Temperature)   5.731700
17                  (count, Area)        inf
18              (max, WindSpeed)   2.886077
19            (min, Temperature)  21.922276
20                            SA        inf
21                            QL        inf
22                           NSW        inf
23      (mean, SoilWaterContent)  15.713195
24     (variance, Precipitation)   3.603813
25             (mean, WindSpeed)   7.356593
26          (min, Precipitation)   1.611378
27                            VI        inf
28            (max, Temperature)  47.541644
29  (variance, RelativeHumidity)   5.536556
30       (min, RelativeHumidity)  21.060184
31          (max, Precipitation)   4.052888
32           (mean, Temperature)  44.570366
```

5 Train Test Split Without Randomization

[9]:
```
# selects 70% of data
training = historicalWeather_pivot.head(n = 28280)
df2 = historicalWeather_pivot.drop(training.index)

# selects 2/3 of the remaining 30 percent
validation = df2.head(n = 8080)
df3 = df2.drop(validation.index)

# the remaining 1/3 of the remaining 30percent
test = df3

#making sure this worked right:
print(len(training)/len(historicalWeather_pivot)*100, '%', ' training')
print(len(validation)/len(historicalWeather_pivot)*100, '%', ' validation')
print(len(test)/len(historicalWeather_pivot)*100, '%', ' test')

#splitting into subtrain and subtest
sub_train_train = training.head(20200)
sub_train_test = training.drop(sub_train_train.index)

#split into xs and ys
cols = historicalWeather_pivot.columns[:-1]
sub_train_train_xs = sub_train_train[cols] # 50%
```

```
sub_train_train_ys = sub_train_train['fire01'] # 50%
sub_train_test_xs = sub_train_test[cols] # 20%
sub_train_test_ys = sub_train_test['fire01'] # 20%
train_xs = training[cols] # 70%
train_ys = training['fire01'] # 70%

#split into validation xs and ys
val_test_xs = validation[cols]
val_test_ys = validation['fire01']

#create the 90% train
ninety_train_xs = pd.concat([training, validation])[cols]
ninety_train_ys = pd.concat([training, validation])['fire01']

#split the test data
test_xs = test[cols]
test_ys = test['fire01']
```

```
69.99306999307 %  training
19.99801999802 %  validation
10.008910008910009 %  test
```

6 Multinomial Logistic Regression

6.1 Hyper Parameter Tuning on the Training Set

[11]:
```
model = LogisticRegression()

model.fit(sub_train_train_xs, sub_train_train_ys)

model.score(sub_train_test_xs, sub_train_test_ys)
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

[11]: 0.7727722772277228

[12]:
```
predictions = model.predict(sub_train_test_xs)
summary = classification_report(sub_train_test_ys, predictions)
```

```
print(summary)
```

```
              precision    recall  f1-score   support

           0       0.69      0.61      0.65      2767
           1       0.81      0.86      0.83      5313

    accuracy                           0.77      8080
   macro avg       0.75      0.73      0.74      8080
weighted avg       0.77      0.77      0.77      8080
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

7 XGBoost

7.1 Hyper Parameter Tuning on the Training Set

[13]:
```
model = XGBClassifier(max_depth=3)
model.fit(sub_train_train_xs, sub_train_train_ys)
print(model.score(sub_train_test_xs, sub_train_test_ys))

predictions = model.predict(sub_train_test_xs)
summary = classification_report(sub_train_test_ys, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/sklearn.py:1224:
UserWarning: The use of label encoder in XGBClassifier is deprecated and will be
removed in a future release. To remove this warning, do the following: 1) Pass
option use_label_encoder=False when constructing XGBClassifier object; and 2)
Encode your labels (y) as integers starting with 0, i.e. 0, 1, 2, …,
[num_class - 1].
  warnings.warn(label_encoder_deprecation_msg, UserWarning)
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/data.py:250:
FutureWarning: pandas.Int64Index is deprecated and will be removed from pandas
in a future version. Use pandas.Index with the appropriate dtype instead.
  elif isinstance(data.columns, (pd.Int64Index, pd.RangeIndex)):
```

```
[15:18:25] WARNING: ../src/learner.cc:1115: Starting in XGBoost 1.3.0, the
default evaluation metric used with the objective 'binary:logistic' was changed
from 'error' to 'logloss'. Explicitly set eval_metric if you'd like to restore
the old behavior.
0.8438118811881188
              precision    recall  f1-score   support
```

```
           0       0.76      0.80      0.78      2767
           1       0.89      0.87      0.88      5313

    accuracy                           0.84      8080
   macro avg       0.83      0.83      0.83      8080
weighted avg       0.85      0.84      0.84      8080
```

7.2 Testing on the validation set

[14]:
```
model = XGBClassifier(max_depth=3)
model.fit(train_xs, train_ys)
print(model.score(val_test_xs.values, val_test_ys.values))

y_pred_proba = model.predict_proba(val_test_xs.values)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = model.predict(val_test_xs.values)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/sklearn.py:1224:
UserWarning: The use of label encoder in XGBClassifier is deprecated and will be
removed in a future release. To remove this warning, do the following: 1) Pass
option use_label_encoder=False when constructing XGBClassifier object; and 2)
Encode your labels (y) as integers starting with 0, i.e. 0, 1, 2, …,
[num_class - 1].
  warnings.warn(label_encoder_deprecation_msg, UserWarning)
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/data.py:250:
FutureWarning: pandas.Int64Index is deprecated and will be removed from pandas
in a future version. Use pandas.Index with the appropriate dtype instead.
  elif isinstance(data.columns, (pd.Int64Index, pd.RangeIndex)):
```

```
[15:18:32] WARNING: ../src/learner.cc:1115: Starting in XGBoost 1.3.0, the
default evaluation metric used with the objective 'binary:logistic' was changed
from 'error' to 'logloss'. Explicitly set eval_metric if you'd like to restore
the old behavior.
0.8497524752475247
AUC:  0.9245560757919986
              precision    recall  f1-score   support

           0       0.75      0.83      0.79      2686
           1       0.91      0.86      0.88      5394

    accuracy                           0.85      8080
   macro avg       0.83      0.84      0.83      8080
```

```
weighted avg       0.86      0.85      0.85      8080
```

8 Multilayer Perceptron

8.0.1 Possible configurations to try:

- [40]
- [40, 40]
- [40, 40, 40]

8.1 Hyper Parameter Tuning

[15]:
```
def find_epochs_and_alpha(train, test, layers):

    for alpha in np.linspace(0.01, 1, 10):
        training_error = []
        test_error = []
        epoch = []
        scalar = StandardScaler()
        cols = train.columns
        xs_train = train[cols[:-1]]
        ys_train = train[cols[-1]]
        xs_test = test[cols[:-1]]
        ys_test = test[cols[-1]]
        train_xs_std = scalar.fit_transform(xs_train)
        test_xs_std = scalar.fit_transform(xs_test)

        for i in tqdm(range(1, 200, 25)):

            mlp = MLPClassifier(hidden_layer_sizes=layers,␣
↪activation='logistic', alpha=alpha,
                                solver='adam', max_iter=i, random_state=777,
                                learning_rate_init=.1, verbose=False)

            mlp.fit(train_xs_std, ys_train)

            #training_error.append(np.sqrt(mean_squared_error(mlp.
↪predict(train_xs_std), ys_train)))
            #test_error.append(np.sqrt(mean_squared_error(mlp.
↪predict(test_xs_std), ys_test)))

            training_error.append(1 - mlp.score(train_xs_std, ys_train))
            test_error.append(1 - mlp.score(test_xs_std, ys_test))

            epoch.append(i)
```

```
        plt.plot(epoch, training_error, label = 'training error')
        plt.plot(epoch, test_error, label = 'test error')
        plt.title(f'training and test error vs epoch number @ ={alpha} and␣
↪{layers}')
        plt.legend()
        plt.show()
```

8.1.1 single layer; 40 neurons

[15]:
```
find_epochs_and_alpha(sub_train_train, sub_train_test, [40])
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:13<00:00,  1.66s/it]
```

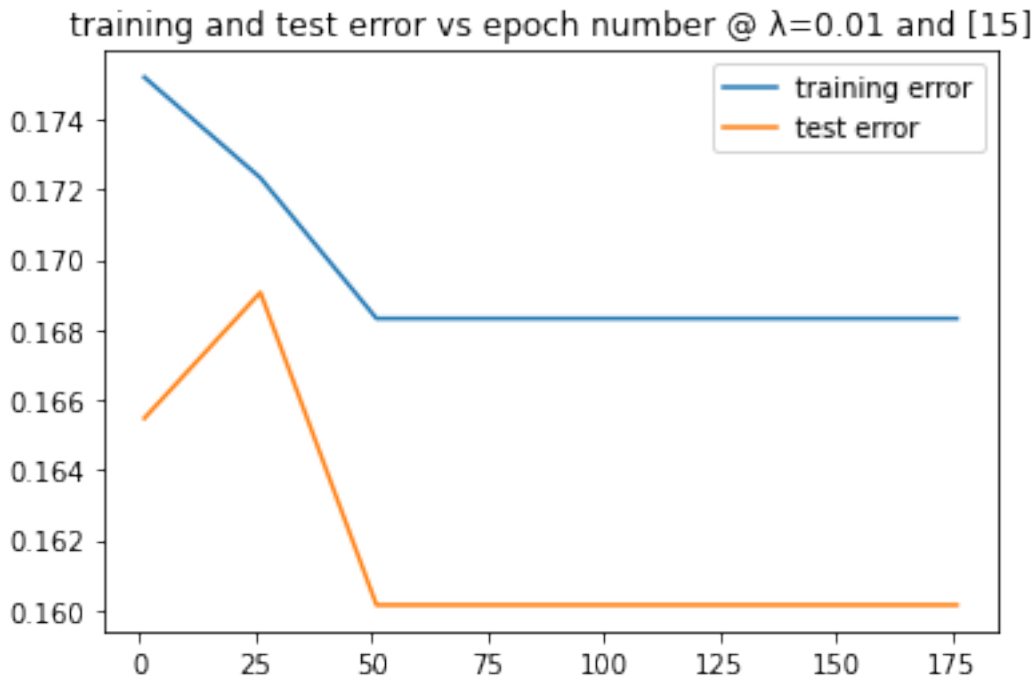


```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
```
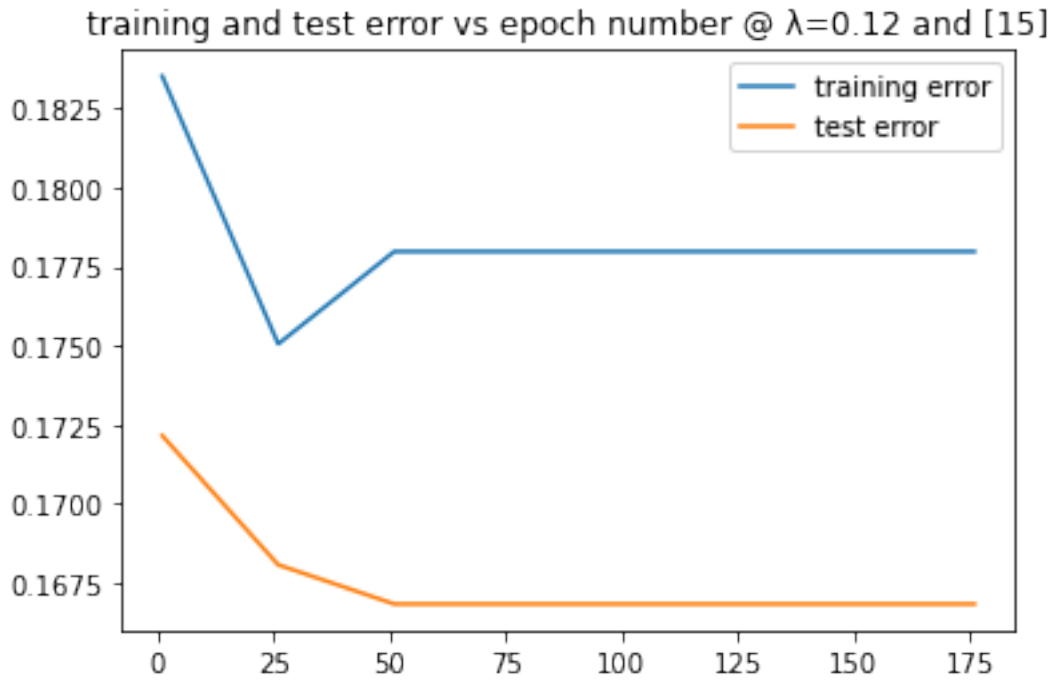

```
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:10<00:00,  1.36s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
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/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
```
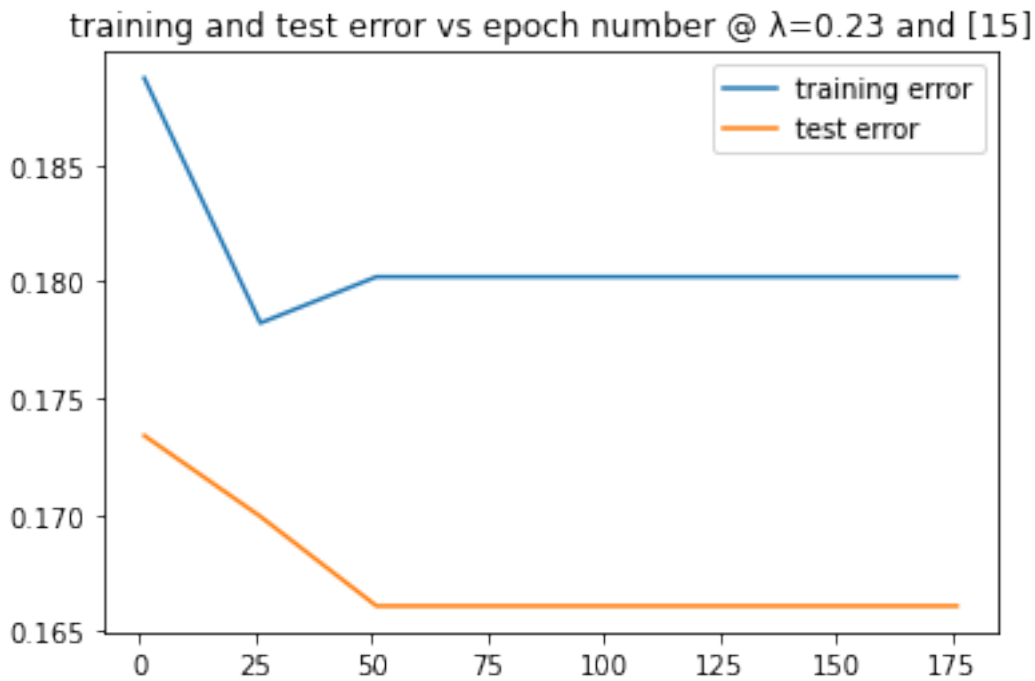

```
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:13<00:00,  1.74s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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```

```
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/home/charlie/anaconda3/lib/python3.9/site-
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/home/charlie/anaconda3/lib/python3.9/site-
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'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:10<00:00,  1.31s/it]
```

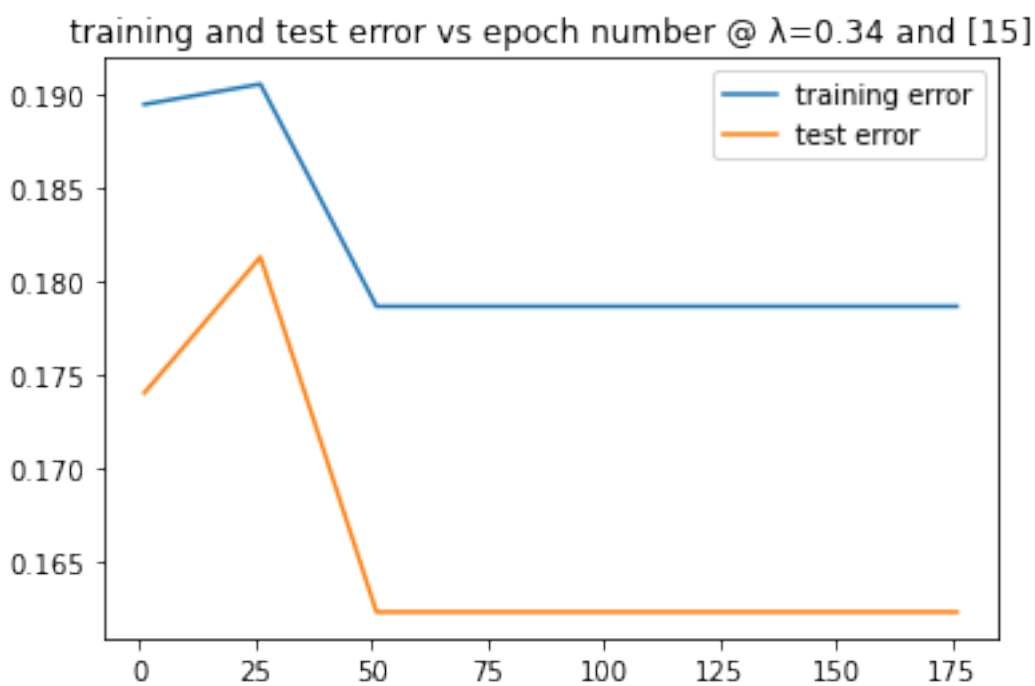


```
/home/charlie/anaconda3/lib/python3.9/site-
```

```
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
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packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:10<00:00,  1.37s/it]
```

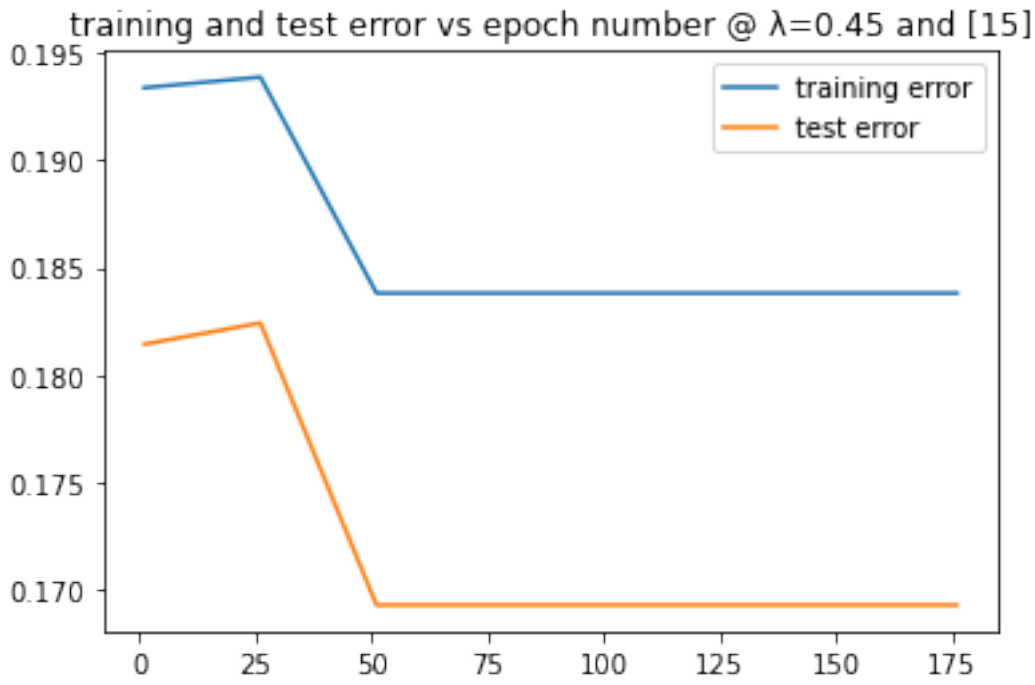


```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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/home/charlie/anaconda3/lib/python3.9/site-
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/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
```
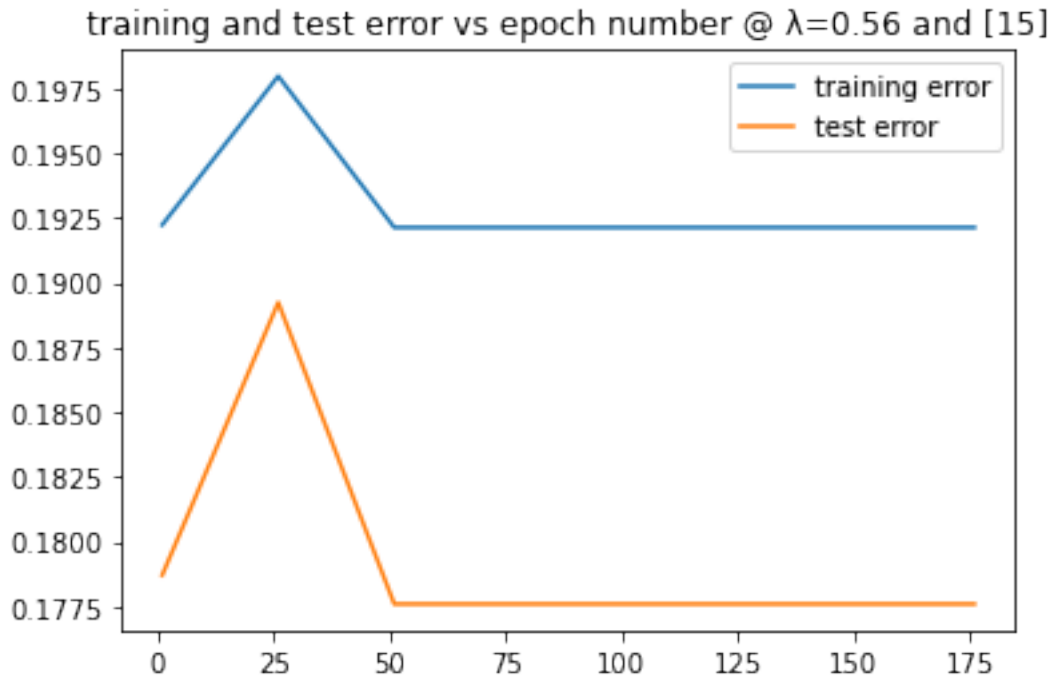

```
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:10<00:00,  1.33s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
```
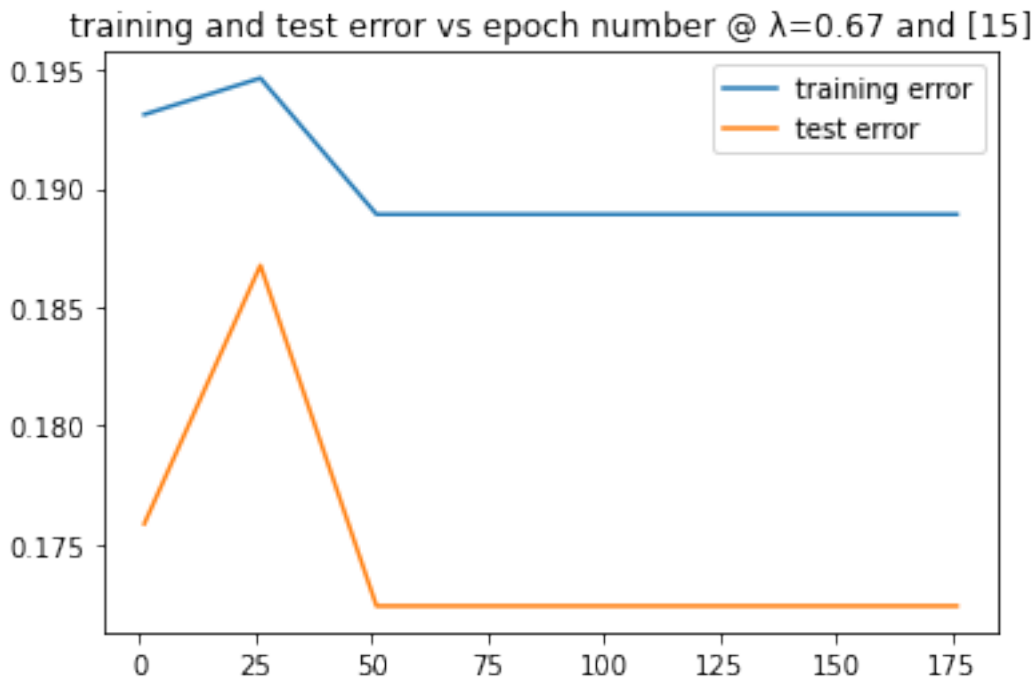

```
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:10<00:00,  1.28s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
```

```
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:13<00:00,  1.65s/it]
```

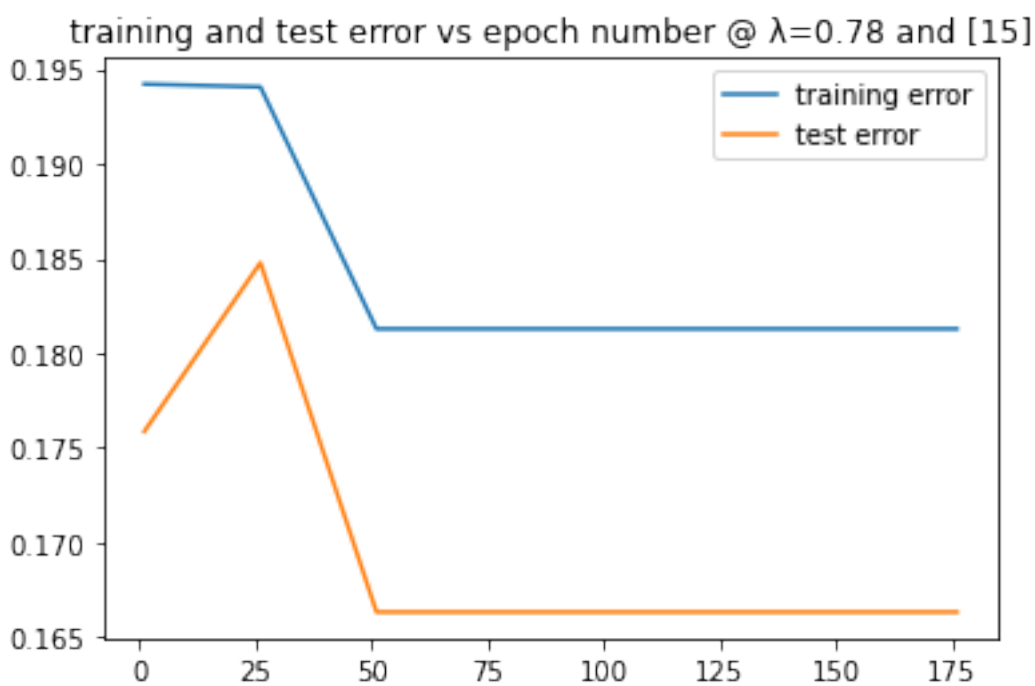


```
/home/charlie/anaconda3/lib/python3.9/site-
```

```
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:05<00:00,  1.55it/s]
```

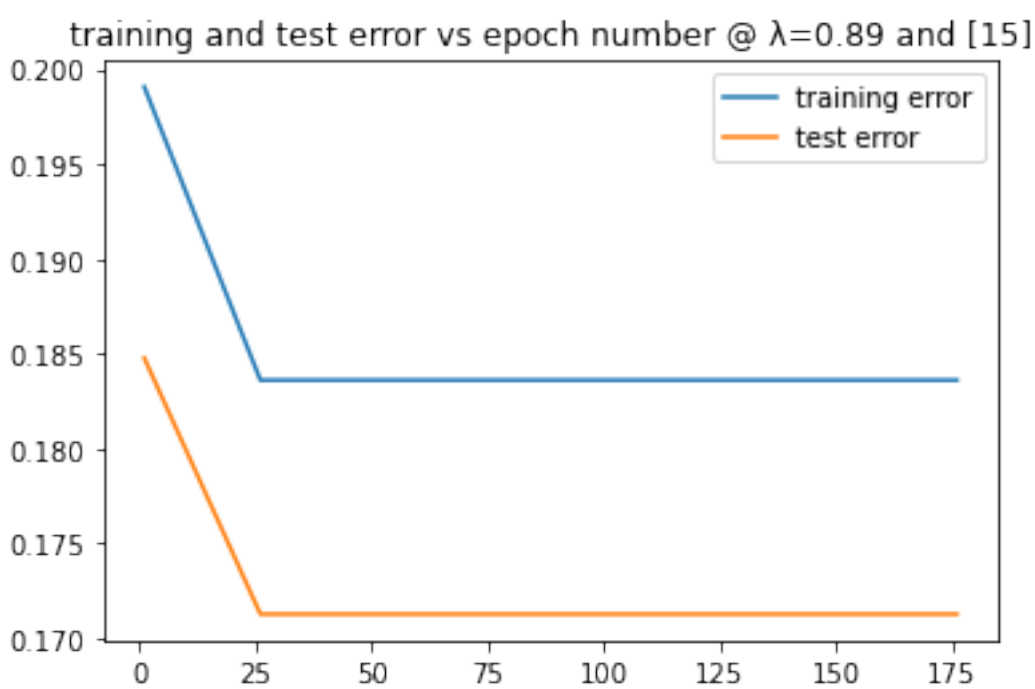

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:10<00:00,  1.33s/it]
```
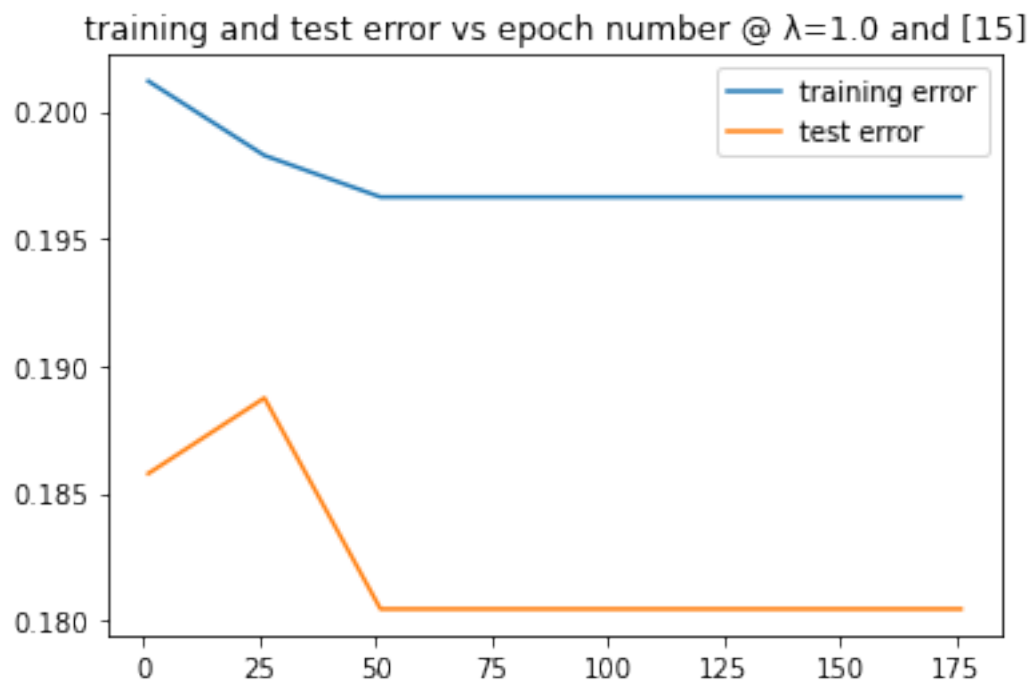



[16]:
```
layers = [40]

scalar = StandardScaler()

xs_train_std = scalar.fit_transform(sub_train_train_xs)
xs_test_std = scalar.fit_transform(sub_train_test_xs)

mlp = MLPClassifier(hidden_layer_sizes=layers, activation='logistic', alpha=.01,
                            solver='adam', max_iter = 100, random_state=777,
                            learning_rate_init=.1, verbose=False)

mlp.fit(xs_train_std, sub_train_train_ys)

print(mlp.score(xs_test_std, sub_train_test_ys))

predictions = mlp.predict(xs_test_std)
summary = classification_report(sub_train_test_ys, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
```

```
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

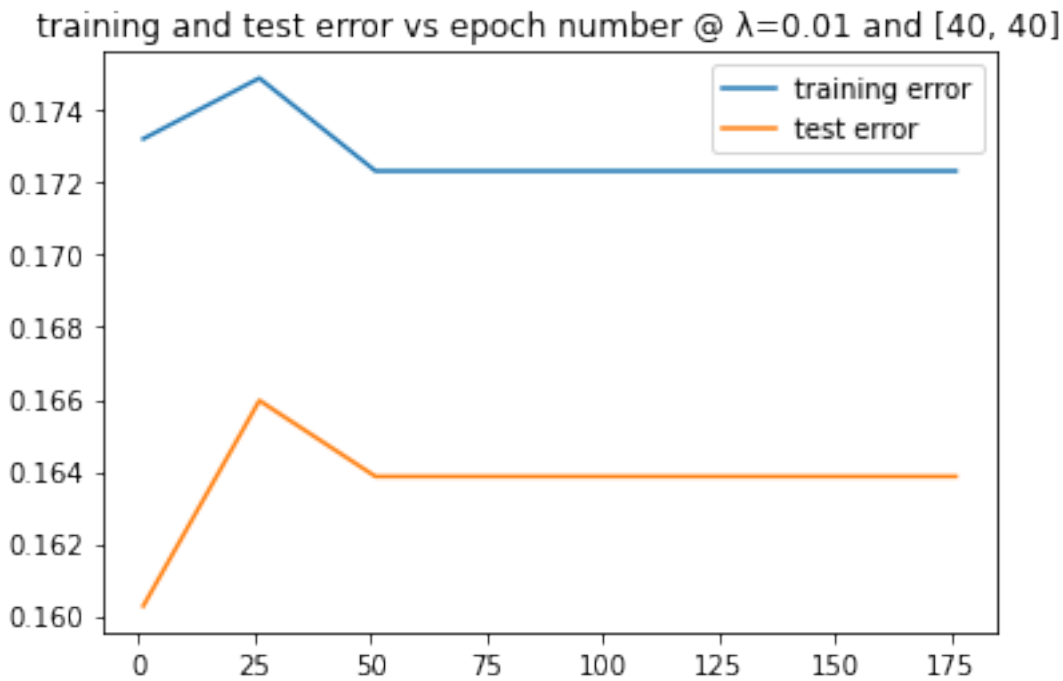
```
0.8349009900990099
              precision    recall  f1-score   support

           0       0.71      0.88      0.79      2767
           1       0.93      0.81      0.87      5313

    accuracy                           0.83      8080
   macro avg       0.82      0.85      0.83      8080
weighted avg       0.85      0.83      0.84      8080
```

8.1.2 double layer, 40 neurons in each

[17]:
```
find_epochs_and_alpha(sub_train_train, sub_train_test, [40, 40])
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
```

```
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                              | 1/8 [00:00<00:01,
5.69it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:24<00:00,  3.05s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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```
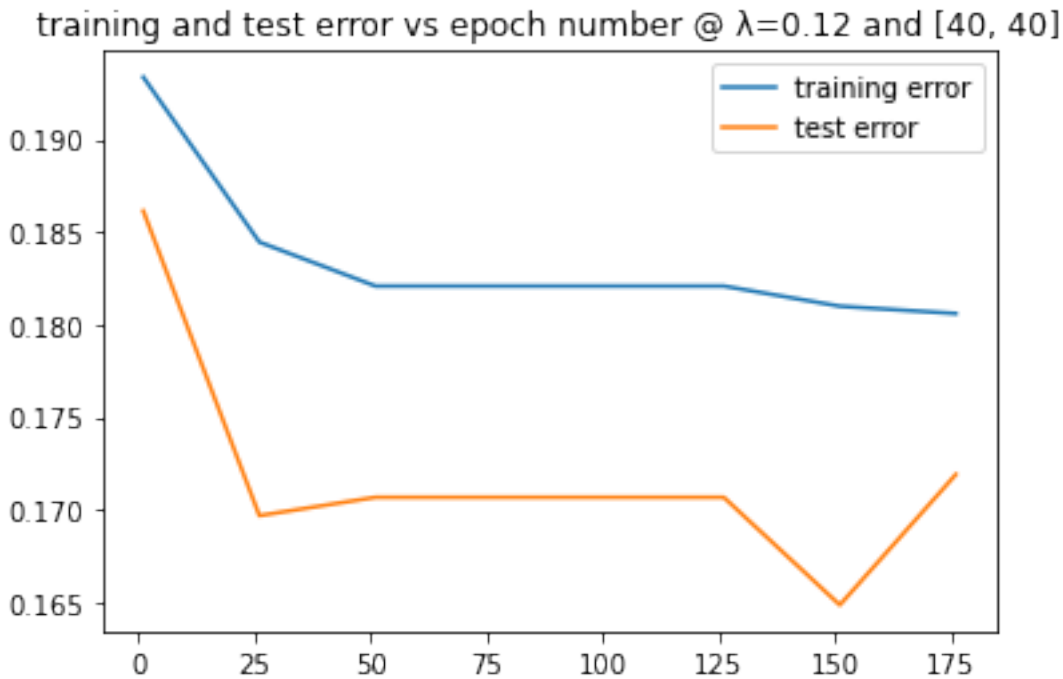

```
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                            | 1/8 [00:00<00:01,
6.45it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
 75%|                              | 6/8 [00:16<00:06,
3.15s/it]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
 88%|                           | 7/8 [00:16<00:02,
2.31s/it]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
100%|                         | 8/8 [00:17<00:00,  2.17s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
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'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
```

```
  warnings.warn(
 12%|                                          | 1/8 [00:00<00:01,
6.05it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
 50%|                                    | 4/8 [00:07<00:09,
2.37s/it]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
 62%|                                 | 5/8 [00:08<00:05,
1.85s/it]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
 75%|                               | 6/8 [00:08<00:02,
1.27s/it]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
 88%|                            | 7/8 [00:08<00:00,
1.11it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:699: UserWarning:
Training interrupted by user.
  warnings.warn("Training interrupted by user.")
100%|                         | 8/8 [00:08<00:00,  1.07s/it]

KeyboardInterrupt
```

[17]:
```
layers = [40, 40]

scalar = StandardScaler()

xs_train_std = scalar.fit_transform(sub_train_train_xs)
xs_test_std = scalar.fit_transform(sub_train_test_xs)

mlp = MLPClassifier(hidden_layer_sizes=layers, activation='logistic', alpha=.01,
                            solver='adam', max_iter = 30, random_state=777,
                            learning_rate_init=.1, verbose=False)

mlp.fit(xs_train_std, sub_train_train_ys)

print(mlp.score(xs_test_std, sub_train_test_ys))

predictions = mlp.predict(xs_test_std)
```

```
summary = classification_report(sub_train_test_ys, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

```
0.8268564356435644
              precision    recall  f1-score   support

           0       0.69      0.90      0.78      2767
           1       0.94      0.79      0.86      5313

    accuracy                           0.83      8080
   macro avg       0.81      0.84      0.82      8080
weighted avg       0.85      0.83      0.83      8080
```

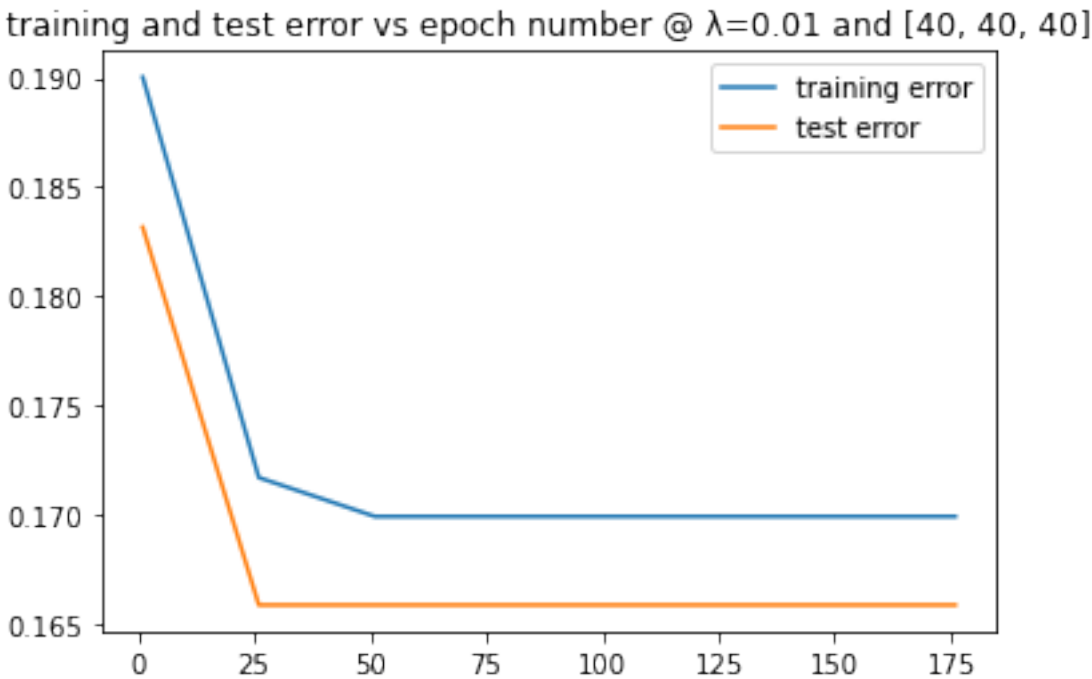
```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (30) reached and
the optimization hasn't converged yet.
  warnings.warn(
```

8.2 Triple layer, 40 neurons each

[18]:
```
find_epochs_and_alpha(sub_train_train, sub_train_test, [40, 40, 40])
```

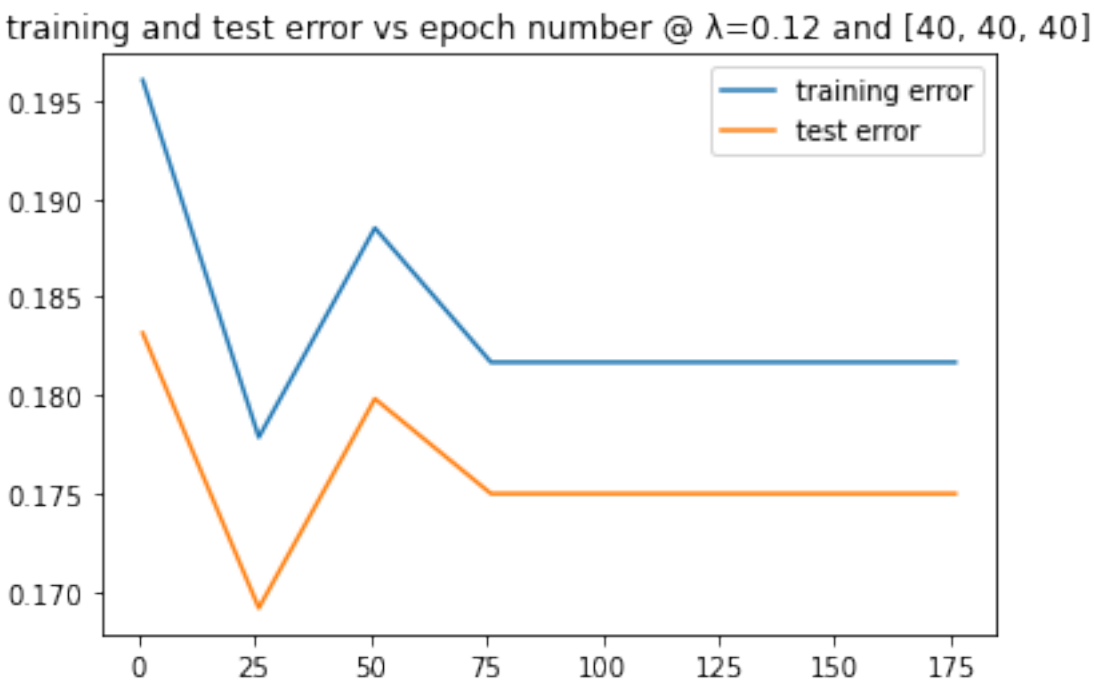
```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
```

```
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                          | 1/8 [00:00<00:01,
6.14it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:24<00:00,  3.05s/it]
```



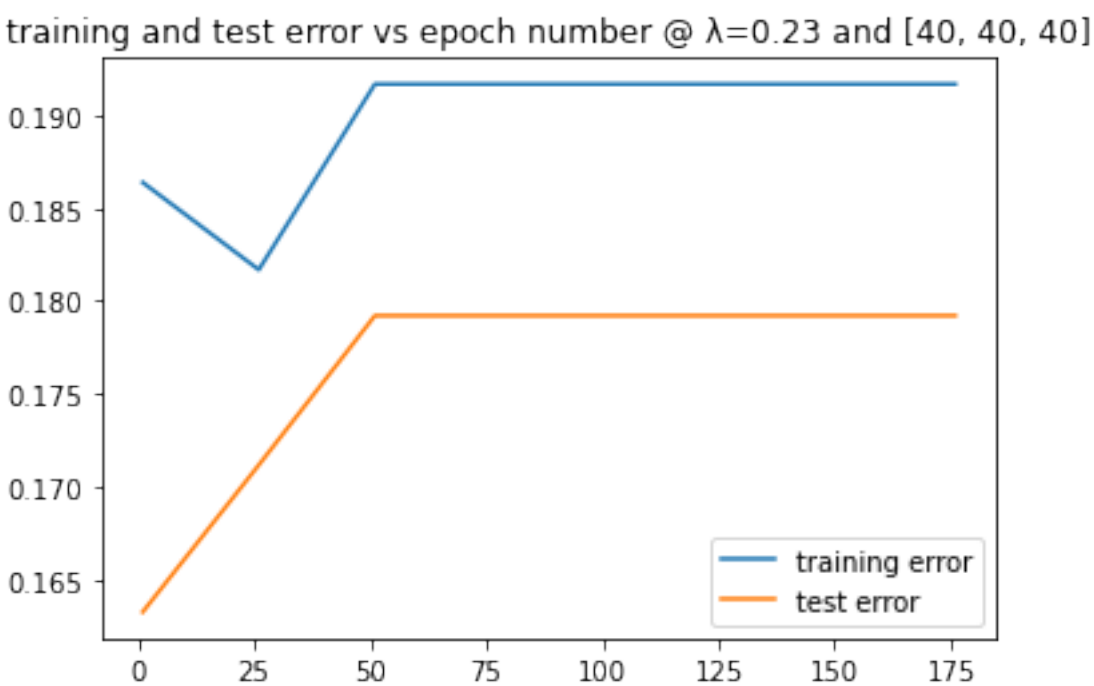

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
```

```
  warnings.warn(
 12%|                                          | 1/8 [00:00<00:01,
5.63it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
 25%|                                       | 2/8 [00:03<00:10,
1.82s/it]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (51) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                         | 8/8 [00:38<00:00,  4.87s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                            | 1/8 [00:00<00:01,
4.61it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:21<00:00,  2.71s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
```

```
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
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  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                          | 1/8 [00:00<00:01,
6.08it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:20<00:00,  2.61s/it]
```
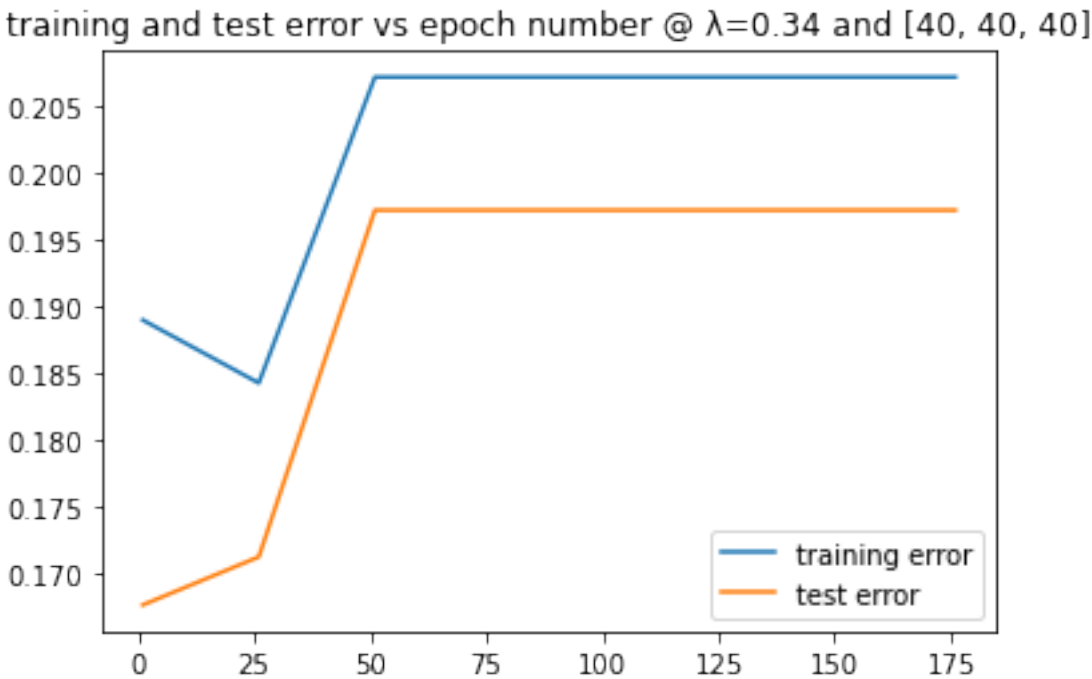
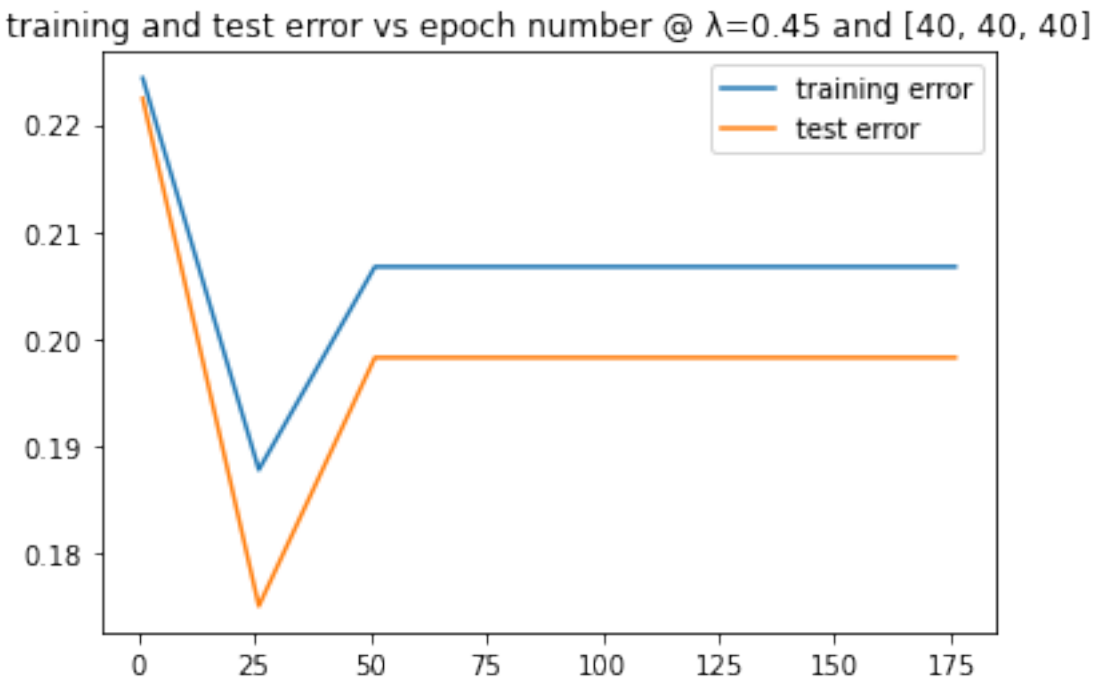



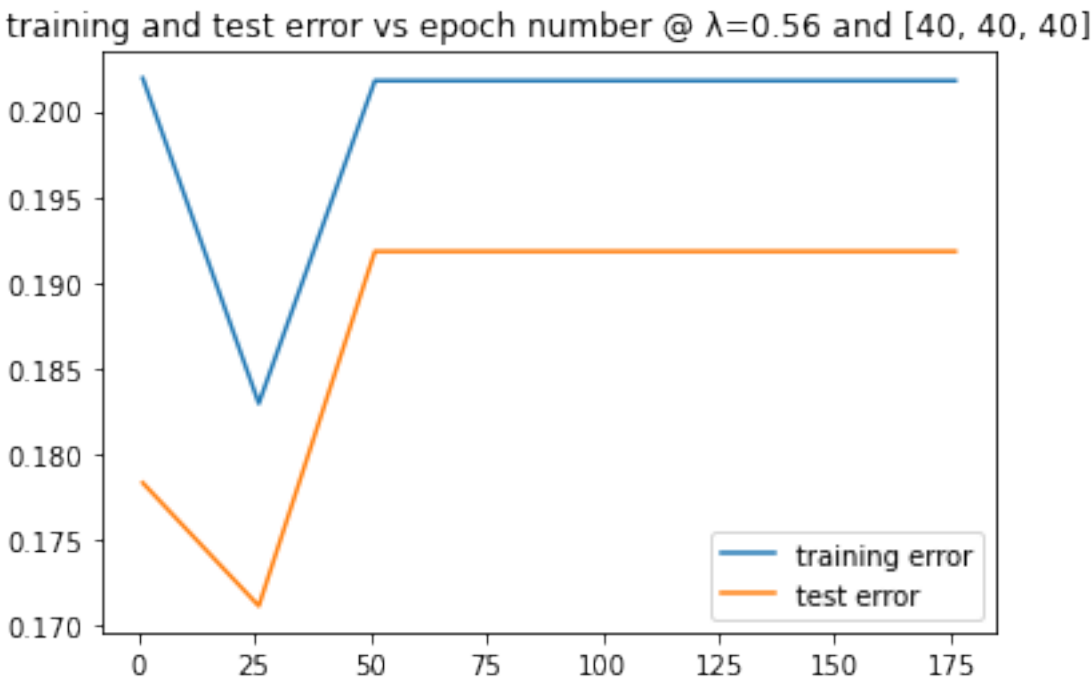

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
```

```
  warnings.warn(
 12%|                                          | 1/8 [00:00<00:01,
6.30it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:20<00:00,  2.60s/it]
```



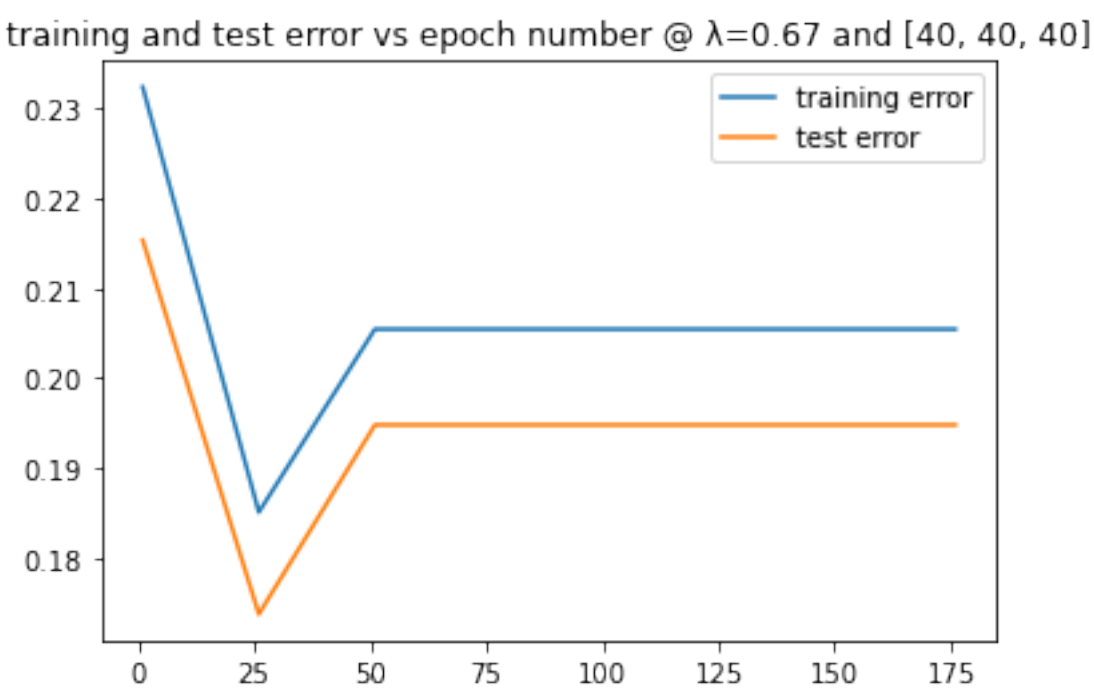

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
```

```
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                            | 1/8 [00:00<00:01,
5.85it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:21<00:00,  2.63s/it]
```

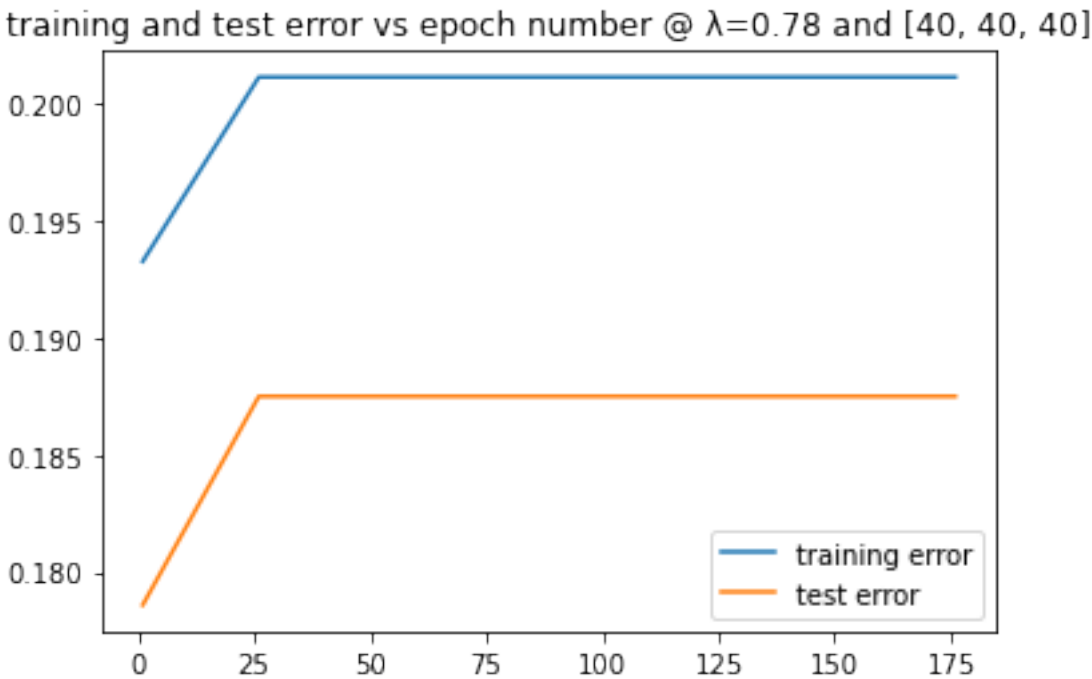



```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
```

```
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
 12%|                                            | 1/8 [00:00<00:01,
5.67it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (26) reached and
the optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:20<00:00,  2.62s/it]
```

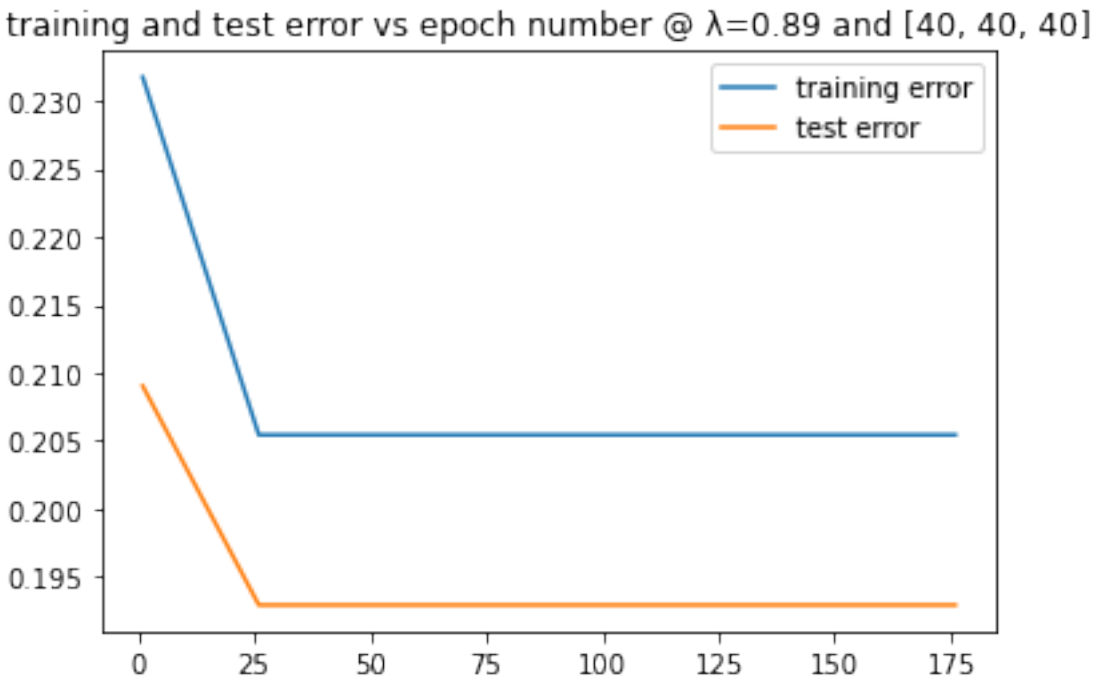


```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
100%|                        | 8/8 [00:19<00:00,  2.45s/it]
```



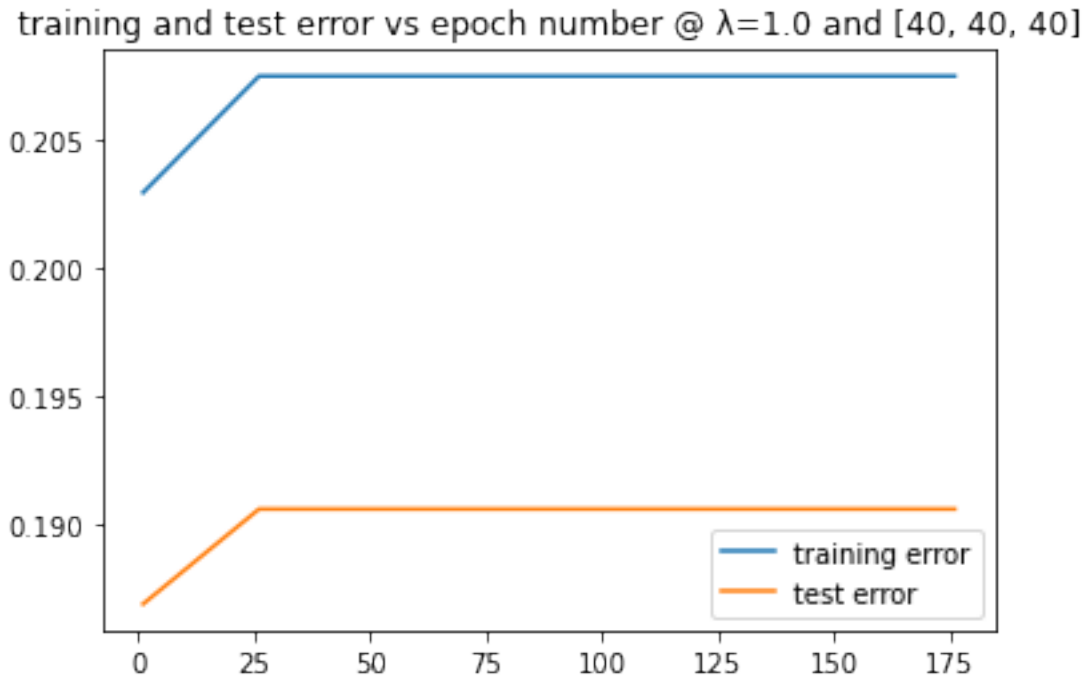

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                          | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
```

```
  warnings.warn(
100%|                          | 8/8 [00:19<00:00,  2.42s/it]
```




```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
  0%|                                                    | 0/8 [00:00<?,
?it/s]/home/charlie/anaconda3/lib/python3.9/site-
```

```
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the
optimization hasn't converged yet.
  warnings.warn(
100%|                          | 8/8 [00:19<00:00,  2.44s/it]
```



[38]:
```
layers = [40, 40, 40]

scalar = StandardScaler()

xs_train_std = scalar.fit_transform(sub_train_train_xs)
xs_test_std = scalar.fit_transform(sub_train_test_xs)

mlp = MLPClassifier(hidden_layer_sizes=layers, activation='logistic', alpha=.01,
                            solver='adam', max_iter = 30, random_state=777,
                            learning_rate_init=.1, verbose=False)

mlp.fit(xs_train_std, sub_train_train_ys)

print(mlp.score(xs_test_std, sub_train_test_ys))

predictions = mlp.predict(xs_test_std)
summary = classification_report(sub_train_test_ys, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

```
0.8362623762376238
              precision    recall  f1-score   support

           0       0.76      0.77      0.76      2767
           1       0.88      0.87      0.88      5313

    accuracy                           0.84      8080
   macro avg       0.82      0.82      0.82      8080
weighted avg       0.84      0.84      0.84      8080
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (30) reached and
the optimization hasn't converged yet.
  warnings.warn(
```

8.3 Testing on the Validation Set

[19]:
```
layers = [40, 40, 40]

scalar = StandardScaler()

xs_train_std = scalar.fit_transform(train_xs.values)
xs_test_std = scalar.fit_transform(val_test_xs.values)

mlp = MLPClassifier(hidden_layer_sizes=layers, activation='logistic', alpha=.01,
```

```
                            solver='adam', max_iter = 30, random_state=777,
                            learning_rate_init=.1, verbose=False)

mlp.fit(xs_train_std, train_ys)

print('SCORE: ', mlp.score(xs_test_std, val_test_ys.values))

y_pred_proba = mlp.predict_proba(val_test_xs)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = mlp.predict(xs_test_std)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
SCORE:  0.8431930693069307
AUC:  0.18478064759083962
              precision    recall  f1-score   support

           0       0.73      0.85      0.78      2686
           1       0.92      0.84      0.88      5394

    accuracy                           0.84      8080
   macro avg       0.82      0.84      0.83      8080
weighted avg       0.85      0.84      0.85      8080


/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:692:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (30) reached and
the optimization hasn't converged yet.
  warnings.warn(
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

9 KNN

9.1 Hyper-Parameter Tuning

[21]:
```
scaler = StandardScaler()
xs_train_std = scaler.fit_transform(sub_train_train_xs.values)
xs_test_std = scaler.fit_transform(sub_train_test_xs.values)

for i in range(1, 33):
```

```
    knn = KNeighborsClassifier(n_neighbors=i)
    knn.fit(xs_train_std, sub_train_train_ys)
    print(i, np.sqrt(mean_squared_error(sub_train_test_ys.values, knn.
↪predict(xs_test_std))))
```

```
1 0.4592287982138216
2 0.46525304516961946
3 0.4412235463138637
4 0.4382684199446637
5 0.42580732871583926
6 0.42769240357903004
7 0.4199540286014814
8 0.4156588471664266
9 0.41326998343446236
10 0.41297040384928985
11 0.41252062654700394
12 0.4122205023700998
13 0.4092071542012737
14 0.41192015952339867
15 0.41026434366535963
16 0.4104151484282786
17 0.41101781416911415
18 0.4096605694696743
19 0.4078438839352214
20 0.4092071542012737
21 0.4081472263973401
22 0.4104151484282786
23 0.4064760414553701
24 0.40708453886407703
25 0.40341978725946903
26 0.4044921029960148
27 0.4035731496627506
28 0.4041860173653504
29 0.4044921029960148
30 0.4044921029960148
31 0.4051035804471359
32 0.40525630564782916
```

9.2 Testing on Validation Set

[23]:
```
scaler = StandardScaler()
xs_train_std = scaler.fit_transform(train_xs.values)
xs_test_std = scaler.fit_transform(val_test_xs.values)

knn = KNeighborsClassifier(n_neighbors=25)
knn.fit(xs_train_std, train_ys.values)
print('SCORE: ', knn.score(xs_test_std, val_test_ys.values))
```

```
y_pred_proba = knn.predict_proba(val_test_xs.values)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = knn.predict(xs_test_std)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
SCORE:  0.842079207920792
AUC:  0.7155256619762562
              precision    recall  f1-score   support

           0       0.73      0.84      0.78      2686
           1       0.91      0.85      0.88      5394

    accuracy                           0.84      8080
   macro avg       0.82      0.84      0.83      8080
weighted avg       0.85      0.84      0.84      8080
```

10 RF

10.1 Hyper Parameter Tuning

[24]:
```
classifier = RandomForestClassifier()
classifier.fit(sub_train_train_xs.values, sub_train_train_ys.values)

print(classifier.score(sub_train_test_xs.values, sub_train_test_ys.values))

predictions = classifier.predict(sub_train_test_xs.values)
summary = classification_report(sub_train_test_ys.values, predictions)
print(summary)
```

```
0.8451732673267327
              precision    recall  f1-score   support

           0       0.75      0.82      0.78      2767
           1       0.90      0.86      0.88      5313

    accuracy                           0.85      8080
   macro avg       0.83      0.84      0.83      8080
weighted avg       0.85      0.85      0.85      8080
```

10.2 Testing on the Validation Set

[25]:
```
classifier = RandomForestClassifier()
classifier.fit(train_xs.values,train_ys.values)

print('SCORE: ', classifier.score(val_test_xs.values, val_test_ys.values))

y_pred_proba = classifier.predict_proba(val_test_xs.values)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = classifier.predict(val_test_xs.values)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
SCORE:  0.8549504950495049
AUC:  0.9252385582723254
              precision    recall  f1-score   support

           0       0.75      0.85      0.80      2686
           1       0.92      0.86      0.89      5394

    accuracy                           0.85      8080
   macro avg       0.83      0.85      0.84      8080
weighted avg       0.86      0.85      0.86      8080
```

11 Logistic GAM

11.1 Hyper Parameter Tuning

[11]:
```
logGAM1 = LogisticGAM() #unfortunately tuning takes too long and kills the␣
↪kernel due to 26 parameters
logGAM1.fit(sub_train_test_xs, sub_train_test_ys)
print('SCORE: ', logGAM1.accuracy(sub_train_test_xs, sub_train_test_ys))

predictions = logGAM1.predict(sub_train_test_xs)
summary = classification_report(sub_train_test_ys, predictions)
print(summary)
```

```
SCORE:  0.8620049504950495
              precision    recall  f1-score   support

           0       0.79      0.80      0.80      2767
           1       0.90      0.89      0.89      5313

    accuracy                           0.86      8080
   macro avg       0.85      0.85      0.85      8080
```

```
weighted avg       0.86      0.86      0.86      8080
```

[14]:
```
logGAM1 = LogisticGAM()
logGAM1.fit(train_xs, train_ys)

print('SCORE: ', logGAM1.accuracy(val_test_xs, val_test_ys))

y_pred_proba = logGAM1.predict_proba(val_test_xs.values)
print('here')
auc = metrics.roc_auc_score(val_test_ys, y_pred_proba)
print('AUC: ', auc)

predictions = logGAM1.predict(val_test_xs)
summary = classification_report(val_test_ys, predictions)
print(summary)
```

```
SCORE:  0.8419554455445545
here
AUC:  0.922498137115479
              precision    recall  f1-score   support

           0       0.74      0.81      0.77      2686
           1       0.90      0.86      0.88      5394

    accuracy                           0.84      8080
   macro avg       0.82      0.83      0.83      8080
weighted avg       0.85      0.84      0.84      8080
```

12 Trees

12.1 Hyper Parameter Tuning

[10]:
```
clf = tree.DecisionTreeClassifier(max_depth = 7)
clf.fit(sub_train_train_xs.values, sub_train_train_ys.values)


print('SCORE: ', clf.score(sub_train_test_xs.values, sub_train_test_ys.values))

predictions = clf.predict(sub_train_test_xs.values)
summary = classification_report(sub_train_test_ys.values, predictions)
print(summary)
```

```
SCORE:  0.823019801980198
              precision    recall  f1-score   support

           0       0.74      0.75      0.75      2767
```

```
           1       0.87      0.86      0.86      5313

    accuracy                           0.82      8080
   macro avg       0.80      0.81      0.80      8080
weighted avg       0.82      0.82      0.82      8080
```

12.2 Testing on the Validation Set

[29]:
```
clf = tree.DecisionTreeClassifier(max_depth = 7)
clf.fit(train_xs.values,train_ys.values)

print('SCORE: ', clf.score(val_test_xs.values, val_test_ys.values))

y_pred_proba = clf.predict_proba(val_test_xs.values)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = clf.predict(val_test_xs)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
SCORE:  0.8330445544554456
AUC:  0.9097452120623809
              precision    recall  f1-score   support

           0       0.72      0.81      0.76      2686
           1       0.90      0.85      0.87      5394

    accuracy                           0.83      8080
   macro avg       0.81      0.83      0.82      8080
weighted avg       0.84      0.83      0.84      8080


/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
```

13 SVC

13.1 Hyper Parameter Tuning

[9]:
```
#careful this takes 1.5 hours! Best parameters arre printed in the following␣
 ↪cell
clf = SVC()
```

```python
param_grid = {'C': [0.1, 1, 10, 100, 1000],
              'gamma': [1, 0.1, 0.01, 0.001, 0.0001],
              'kernel': ['rbf']}

gridrbf = GridSearchCV(SVC(), param_grid, refit = True, verbose = 3)

gridrbf.fit(sub_train_train_xs.values, sub_train_train_ys.values)
```

```
Fitting 5 folds for each of 25 candidates, totalling 125 fits
```

```
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
Input In [9], in <cell line: 9>()
      3 param_grid = {'C': [0.1, 1, 10, 100, 1000],
      4               'gamma': [1, 0.1, 0.01, 0.001, 0.0001],
      5               'kernel': ['rbf']}
      7 gridrbf = GridSearchCV(SVC(), param_grid, refit = True, verbose = 3)
----> 9 gridrbf.fit(sub_train_train_xs.values, sub_train_train_ys.values)

File ~/anaconda3/lib/python3.9/site-packages/sklearn/model_selection/_search.py
 ↪891, in BaseSearchCV.fit(self, X, y, groups, **fit_params)
    885     results = self._format_results(
    886         all_candidate_params, n_splits, all_out, all_more_results
    887     )
    889     return results
--> 891 self._run_search(evaluate_candidates)
    893 # multimetric is determined here because in the case of a callable
    894 # self.scoring the return type is only known after calling
    895 first_test_score = all_out[0]["test_scores"]

File ~/anaconda3/lib/python3.9/site-packages/sklearn/model_selection/_search.py
 ↪1392, in GridSearchCV._run_search(self, evaluate_candidates)
   1390 def _run_search(self, evaluate_candidates):
   1391     """Search all candidates in param_grid"""
-> 1392     evaluate_candidates(ParameterGrid(self.param_grid))

File ~/anaconda3/lib/python3.9/site-packages/sklearn/model_selection/_search.py
 ↪838, in BaseSearchCV.fit.<locals>.evaluate_candidates(candidate_params, cv,␣
 ↪more_results)
    830 if self.verbose > 0:
    831     print(
    832         "Fitting {0} folds for each of {1} candidates,"
    833         " totalling {2} fits".format(
    834             n_splits, n_candidates, n_candidates * n_splits
    835         )
    836     )
--> 838 out = parallel(
```

```
    839     delayed(_fit_and_score)(
    840         clone(base_estimator),
    841         X,
    842         y,
    843         train=train,
    844         test=test,
    845         parameters=parameters,
    846         split_progress=(split_idx, n_splits),
    847         candidate_progress=(cand_idx, n_candidates),
    848         **fit_and_score_kwargs,
    849     )
    850     for (cand_idx, parameters), (split_idx, (train, test)) in product(
    851         enumerate(candidate_params), enumerate(cv.split(X, y, groups))
    852     )
    853 )
    855 if len(out) < 1:
    856     raise ValueError(
    857         "No fits were performed. "
    858         "Was the CV iterator empty? "
    859         "Were there no candidates?"
    860     )

File ~/anaconda3/lib/python3.9/site-packages/joblib/parallel.py:1043, in␣
 ↪Parallel.__call__(self, iterable)
   1034 try:
   1035     # Only set self._iterating to True if at least a batch
   1036     # was dispatched. In particular this covers the edge
   (…)
   1040     # was very quick and its callback already dispatched all the
   1041     # remaining jobs.
   1042     self._iterating = False
-> 1043     if self.dispatch_one_batch(iterator):
   1044         self._iterating = self._original_iterator is not None
   1046     while self.dispatch_one_batch(iterator):

File ~/anaconda3/lib/python3.9/site-packages/joblib/parallel.py:861, in Parallel.
 ↪dispatch_one_batch(self, iterator)
    859     return False
    860 else:
--> 861     self._dispatch(tasks)
    862     return True

File ~/anaconda3/lib/python3.9/site-packages/joblib/parallel.py:779, in Parallel.
 ↪_dispatch(self, batch)
    777 with self._lock:
    778     job_idx = len(self._jobs)
--> 779     job = self._backend.apply_async(batch, callback=cb)
    780     # A job can complete so quickly than its callback is
```

```
    781     # called before we get here, causing self._jobs to
    782     # grow. To ensure correct results ordering, .insert is
    783     # used (rather than .append) in the following line
    784     self._jobs.insert(job_idx, job)

File ~/anaconda3/lib/python3.9/site-packages/joblib/_parallel_backends.py:208,␣
↪in SequentialBackend.apply_async(self, func, callback)
    206 def apply_async(self, func, callback=None):
    207     """Schedule a func to be run"""
--> 208     result = ImmediateResult(func)
    209     if callback:
    210         callback(result)

File ~/anaconda3/lib/python3.9/site-packages/joblib/_parallel_backends.py:572,␣
↪in ImmediateResult.__init__(self, batch)
    569 def __init__(self, batch):
    570     # Don't delay the application, to avoid keeping the input
    571     # arguments in memory
--> 572     self.results = batch()

File ~/anaconda3/lib/python3.9/site-packages/joblib/parallel.py:262, in␣
↪BatchedCalls.__call__(self)
    258 def __call__(self):
    259     # Set the default nested backend to self._backend but do not set th
    260     # change the default number of processes to -1
    261     with parallel_backend(self._backend, n_jobs=self._n_jobs):
--> 262         return [func(*args, **kwargs)
    263                 for func, args, kwargs in self.items]

File ~/anaconda3/lib/python3.9/site-packages/joblib/parallel.py:262, in␣
↪<listcomp>(.0)
    258 def __call__(self):
    259     # Set the default nested backend to self._backend but do not set th
    260     # change the default number of processes to -1
    261     with parallel_backend(self._backend, n_jobs=self._n_jobs):
--> 262         return [func(*args, **kwargs)
    263                 for func, args, kwargs in self.items]

File ~/anaconda3/lib/python3.9/site-packages/sklearn/utils/fixes.py:216, in␣
↪_FuncWrapper.__call__(self, *args, **kwargs)
    214 def __call__(self, *args, **kwargs):
    215     with config_context(**self.config):
--> 216         return self.function(*args, **kwargs)

File ~/anaconda3/lib/python3.9/site-packages/sklearn/model_selection/_validatio
↪py:680, in _fit_and_score(estimator, X, y, scorer, train, test, verbose,␣
↪parameters, fit_params, return_train_score, return_parameters,␣
↪return_n_test_samples, return_times, return_estimator, split_progress,␣
↪candidate_progress, error_score)
```

```
    678         estimator.fit(X_train, **fit_params)
    679     else:
--> 680         estimator.fit(X_train, y_train, **fit_params)
    682 except Exception:
    683     # Note fit time as time until error
    684     fit_time = time.time() - start_time

File ~/anaconda3/lib/python3.9/site-packages/sklearn/svm/_base.py:255, in␣
 ↪BaseLibSVM.fit(self, X, y, sample_weight)
    252     print("[LibSVM]", end="")
    254 seed = rnd.randint(np.iinfo("i").max)
--> 255 fit(X, y, sample_weight, solver_type, kernel, random_seed=seed)
    256 # see comment on the other call to np.iinfo in this file
    258 self.shape_fit_ = X.shape if hasattr(X, "shape") else (n_samples,)

File ~/anaconda3/lib/python3.9/site-packages/sklearn/svm/_base.py:315, in␣
 ↪BaseLibSVM._dense_fit(self, X, y, sample_weight, solver_type, kernel,␣
 ↪random_seed)
    302 libsvm.set_verbosity_wrap(self.verbose)
    304 # we don't pass **self.get_params() to allow subclasses to
    305 # add other parameters to __init__
    306 (
    307     self.support_,
    308     self.support_vectors_,
    309     self._n_support,
    310     self.dual_coef_,
    311     self.intercept_,
    312     self._probA,
    313     self._probB,
    314     self.fit_status_,
--> 315 ) = libsvm.fit(
    316     X,
    317     y,
    318     svm_type=solver_type,
    319     sample_weight=sample_weight,
    320     class_weight=self.class_weight_,
    321     kernel=kernel,
    322     C=self.C,
    323     nu=self.nu,
    324     probability=self.probability,
    325     degree=self.degree,
    326     shrinking=self.shrinking,
    327     tol=self.tol,
    328     cache_size=self.cache_size,
    329     coef0=self.coef0,
    330     gamma=self._gamma,
    331     epsilon=self.epsilon,
    332     max_iter=self.max_iter,
```

```
    333     random_seed=random_seed,
    334 )
    336 self._warn_from_fit_status()

KeyboardInterrupt:
```

[12]:
```
clf = SVC(C=100,gamma=.0001,kernel='rbf', probability=True)
#best parameters = {'C': 100, 'gamma': 0.0001, 'kernel': 'rbf'}

clf.fit(sub_train_train_xs.values, sub_train_train_ys.values)
print('SCORE: ', clf.score(sub_train_test_xs.values, sub_train_test_ys.values))
predictions = clf.predict(sub_train_test_xs.values)
summary = classification_report(sub_train_test_ys.values, predictions)
print(predictions)
print(summary)
```

```
SCORE:  0.8323019801980198
[0 1 0 … 0 0 1]
              precision    recall  f1-score   support

           0       0.73      0.80      0.77      2767
           1       0.89      0.85      0.87      5313

    accuracy                           0.83      8080
   macro avg       0.81      0.83      0.82      8080
weighted avg       0.84      0.83      0.83      8080
```

14 Testing on Validation Set

[13]:
```
clf.fit(train_xs.values,train_ys.values)

print('SCORE: ', clf.score(val_test_xs.values, val_test_ys.values))

y_pred_proba = clf.predict_proba(val_test_xs.values)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = clf.predict(val_test_xs)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
SCORE:  0.8389851485148515
AUC:  0.8952849074465961
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
```

```
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(
              precision    recall  f1-score   support

           0       0.73      0.82      0.77      2686
           1       0.90      0.85      0.88      5394

    accuracy                           0.84      8080
   macro avg       0.82      0.83      0.82      8080
weighted avg       0.85      0.84      0.84      8080
```

15 Naive Bayes

15.1 Hyper Parameter Tuning

[14]:
```
clf = naive_bayes.MultinomialNB()
scaler = MinMaxScaler()

params = {'alpha': [0.01, 0.1, 0.5, 1.0, 10.0, ],
         }

scaled_xs = scaler.fit_transform(sub_train_train_xs.values)
scaled_test_xs = scaler.fit_transform(sub_train_test_xs.values)


grid = GridSearchCV(estimator=clf, param_grid=params, verbose=1, cv=10,␣
 ↪n_jobs=-1)

grid.fit(scaled_xs, sub_train_train_ys.values)


print('SCORE: ', grid.score(scaled_test_xs, sub_train_test_ys.values))

predictions = grid.predict(scaled_test_xs)
summary = classification_report(sub_train_test_ys.values, predictions)
print(summary)
```

```
Fitting 10 folds for each of 5 candidates, totalling 50 fits
SCORE:  0.808539603960396
              precision    recall  f1-score   support

           0       0.71      0.74      0.72      2767
           1       0.86      0.85      0.85      5313

    accuracy                           0.81      8080
```

```
   macro avg       0.79      0.79      0.79      8080
weighted avg       0.81      0.81      0.81      8080
```

15.2 Testing on the Validation Set

[15]:
```
scaler = MinMaxScaler()
scaled_xs = scaler.fit_transform(train_xs.values)
scaled_test_xs = scaler.fit_transform(val_test_xs.values)

grid.fit(scaled_xs, train_ys.values)

print('SCORE: ', grid.score(scaled_test_xs, val_test_ys.values))

y_pred_proba = grid.predict_proba(scaled_test_xs)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = grid.predict(scaled_test_xs)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
Fitting 10 folds for each of 5 candidates, totalling 50 fits
SCORE:  0.8131188118811881
AUC:  0.8836357708062597
              precision    recall  f1-score   support

           0       0.70      0.76      0.73      2686
           1       0.87      0.84      0.86      5394

    accuracy                           0.81      8080
   macro avg       0.79      0.80      0.79      8080
weighted avg       0.82      0.81      0.81      8080
```

16 XGBoosted Random Forest

16.1 Hyper Parameter Tuning

[61]:
```
classifier = XGBRFClassifier(max_depth=15)
classifier.fit(sub_train_train_xs.values, sub_train_train_ys.values)

print(classifier.score(sub_train_test_xs.values, sub_train_test_ys.values))

predictions = classifier.predict(sub_train_test_xs.values)
summary = classification_report(sub_train_test_ys.values, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/sklearn.py:1224:
UserWarning: The use of label encoder in XGBClassifier is deprecated and will be
removed in a future release. To remove this warning, do the following: 1) Pass
option use_label_encoder=False when constructing XGBClassifier object; and 2)
Encode your labels (y) as integers starting with 0, i.e. 0, 1, 2, …,
[num_class - 1].
  warnings.warn(label_encoder_deprecation_msg, UserWarning)

[19:38:08] WARNING: ../src/learner.cc:1115: Starting in XGBoost 1.3.0, the
default evaluation metric used with the objective 'binary:logistic' was changed
from 'error' to 'logloss'. Explicitly set eval_metric if you'd like to restore
the old behavior.
0.8471534653465347
              precision    recall  f1-score   support

           0       0.75      0.82      0.79      2767
           1       0.90      0.86      0.88      5313

    accuracy                           0.85      8080
   macro avg       0.83      0.84      0.83      8080
weighted avg       0.85      0.85      0.85      8080
```

16.2 Testing on the Validation Set

[62]:
```python
model = XGBRFClassifier(max_depth=15)
model.fit(train_xs, train_ys)
print(model.score(val_test_xs.values, val_test_ys.values))

y_pred_proba = model.predict_proba(val_test_xs.values)[::,1]
auc = metrics.roc_auc_score(val_test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = model.predict(val_test_xs.values)
summary = classification_report(val_test_ys.values, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/sklearn.py:1224:
UserWarning: The use of label encoder in XGBClassifier is deprecated and will be
removed in a future release. To remove this warning, do the following: 1) Pass
option use_label_encoder=False when constructing XGBClassifier object; and 2)
Encode your labels (y) as integers starting with 0, i.e. 0, 1, 2, …,
[num_class - 1].
  warnings.warn(label_encoder_deprecation_msg, UserWarning)
/home/charlie/anaconda3/lib/python3.9/site-packages/xgboost/data.py:250:
FutureWarning: pandas.Int64Index is deprecated and will be removed from pandas
in a future version. Use pandas.Index with the appropriate dtype instead.
  elif isinstance(data.columns, (pd.Int64Index, pd.RangeIndex)):
```

```
[19:38:24] WARNING: ../src/learner.cc:1115: Starting in XGBoost 1.3.0, the
default evaluation metric used with the objective 'binary:logistic' was changed
from 'error' to 'logloss'. Explicitly set eval_metric if you'd like to restore
the old behavior.
0.849009900990099
AUC:  0.9253083042822738
              precision    recall  f1-score   support

           0       0.74      0.83      0.79      2686
           1       0.91      0.86      0.88      5394

    accuracy                           0.85      8080
   macro avg       0.83      0.85      0.83      8080
weighted avg       0.86      0.85      0.85      8080
```

17 The Winner

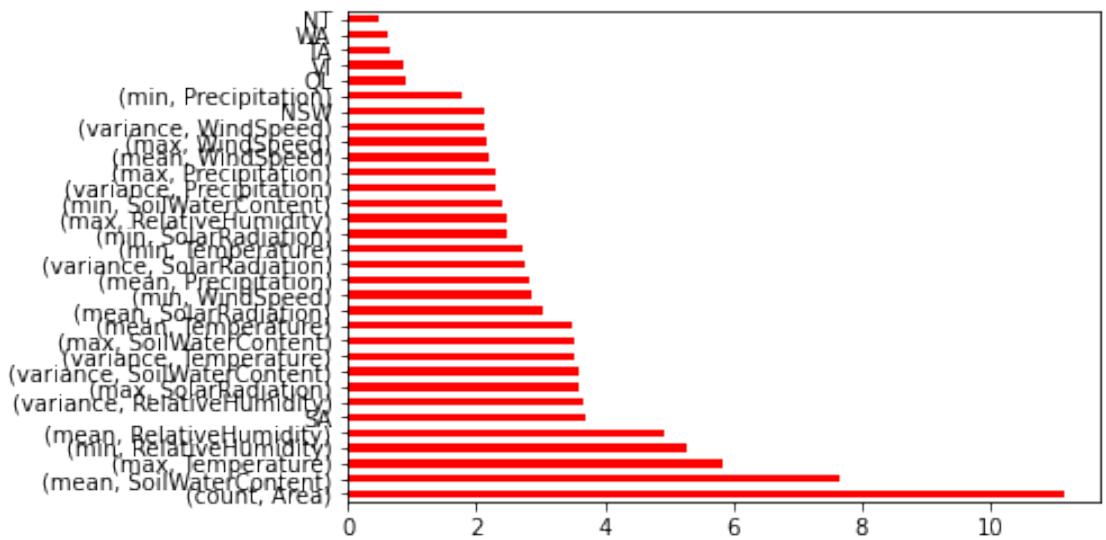
[69]:
```
classifier = RandomForestClassifier()
classifier.fit(ninety_train_xs, ninety_train_ys)

print('SCORE: ', classifier.score(test_xs.values, test_ys.values))

y_pred_proba = classifier.predict_proba(test_xs.values)[::,1]
auc = metrics.roc_auc_score(test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = classifier.predict(test_xs.values)
summary = classification_report(test_ys.values, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['str',
'tuple']. An error will be raised in 1.2.
  warnings.warn(

SCORE:  0.8521266073194856
AUC:  0.9276984440125813
              precision    recall  f1-score   support

           0       0.76      0.83      0.79      1377
           1       0.91      0.87      0.89      2667

    accuracy                           0.85      4044
   macro avg       0.83      0.85      0.84      4044
weighted avg       0.86      0.85      0.85      4044
```

18 Experimentation with Bagging and Importance

[75]:
```
importance = pd.Series(classifier.feature_importances_ * 100, index = cols)
importance = importance.sort_values(axis=0, ascending = False)
importance.plot(kind='barh', color='r', )
```

[75]: <AxesSubplot:>



[13]:
```
cols2 = [('mean', 'RelativeHumidity'), ('min', 'RelativeHumidity'), ('max',␣
↪'Temperature'), ('mean', 'SoilWaterContent'), ('count', 'Area')]

classifier = RandomForestClassifier()
classifier.fit(ninety_train_xs[cols2], ninety_train_ys)

print('SCORE: ', classifier.score(test_xs[cols2].values, test_ys.values))

y_pred_proba = classifier.predict_proba(test_xs[cols2].values)[::,1]
auc = metrics.roc_auc_score(test_ys.values, y_pred_proba)
print('AUC: ', auc)

predictions = classifier.predict(test_xs[cols2].values)
summary = classification_report(test_ys.values, predictions)
print(summary)
```

```
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/utils/validation.py:1688: FutureWarning: Feature names only
support names that are all strings. Got feature names with dtypes: ['tuple']. An
error will be raised in 1.2.
  warnings.warn(
```

```
SCORE:  0.8303659742828883
AUC:  0.9103272493988359
              precision    recall  f1-score   support

           0       0.73      0.80      0.76      1377
           1       0.89      0.84      0.87      2667

    accuracy                           0.83      4044
   macro avg       0.81      0.82      0.82      4044
weighted avg       0.84      0.83      0.83      4044
```

19 Experimentation with LogisticGAM

[27]:
```
data = historicalWeather_pivot

def month_by_month(data):

    logGAM1 = LogisticGAM()

    for i in range(10):
        xs_train = data.iloc[0+30*i:30+30*i][cols]
        ys_train = data.iloc[0+30*i:30+30*i]['fire01']


        xs_test = data.iloc[60+30*i:90+30*i][cols]
        ys_test = data.iloc[60+30*i:90+30*i]['fire01']


        logGAM1 = LogisticGAM()
        logGAM1.fit(xs_train, ys_train)
        print(i)
        print('SCORE: ', logGAM1.accuracy(xs_test, ys_test))

        y_pred_proba = logGAM1.predict_proba(xs_test.values)
        auc = metrics.roc_auc_score(ys_test, y_pred_proba)
        print('AUC: ', auc)

        predictions = logGAM1.predict(xs_test)
        summary = classification_report(ys_test, predictions)
        print(summary)

month_by_month(data)
```

```
0
SCORE:  0.5
AUC:  0.8295454545454546
```

```
              precision    recall  f1-score   support

           0       0.35      1.00      0.52         8
           1       1.00      0.32      0.48        22

    accuracy                           0.50        30
   macro avg       0.67      0.66      0.50        30
weighted avg       0.83      0.50      0.49        30

did not converge
1
SCORE:  0.7666666666666667
AUC:  0.9316770186335405
              precision    recall  f1-score   support

           0       0.00      0.00      0.00         7
           1       0.77      1.00      0.87        23

    accuracy                           0.77        30
   macro avg       0.38      0.50      0.43        30
weighted avg       0.59      0.77      0.67        30


/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
2
SCORE:  0.7666666666666667
AUC:  0.6708074534161491
              precision    recall  f1-score   support

           0       0.00      0.00      0.00         7
           1       0.77      1.00      0.87        23

    accuracy                           0.77        30
   macro avg       0.38      0.50      0.43        30
```

```
weighted avg       0.59      0.77      0.67        30

/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

```
3
SCORE:  0.6666666666666666
AUC:  0.685
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        10
           1       0.67      1.00      0.80        20

    accuracy                           0.67        30
   macro avg       0.33      0.50      0.40        30
weighted avg       0.44      0.67      0.53        30

/home/charlie/anaconda3/lib/python3.9/site-
packages/sklearn/metrics/_classification.py:1318: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with no
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```

```
4
SCORE:  0.8333333333333334
```

```
AUC:  0.7569444444444444
              precision    recall  f1-score   support

           0       1.00      0.17      0.29         6
           1       0.83      1.00      0.91        24

    accuracy                           0.83        30
   macro avg       0.91      0.58      0.60        30
weighted avg       0.86      0.83      0.78        30

5
SCORE:  0.6666666666666666
AUC:  0.6849999999999999
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        10
           1       0.67      1.00      0.80        20

    accuracy                           0.67        30
   macro avg       0.33      0.50      0.40        30
weighted avg       0.44      0.67      0.53        30


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6
SCORE:  0.4666666666666667
AUC:  0.59375
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        16
           1       0.47      1.00      0.64        14

    accuracy                           0.47        30
   macro avg       0.23      0.50      0.32        30
```

```
weighted avg       0.22      0.47      0.30        30

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```

```
7
SCORE:  0.5
AUC:  0.88
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        15
           1       0.50      1.00      0.67        15

    accuracy                           0.50        30
   macro avg       0.25      0.50      0.33        30
weighted avg       0.25      0.50      0.33        30

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  _warn_prf(average, modifier, msg_start, len(result))
```

```
8
SCORE:  0.5
```

```
AUC:  0.8708133971291866
              precision    recall  f1-score   support

           0       0.42      1.00      0.59        11
           1       1.00      0.21      0.35        19

    accuracy                           0.50        30
   macro avg       0.71      0.61      0.47        30
weighted avg       0.79      0.50      0.44        30

did not converge
9
SCORE:  0.8333333333333334
AUC:  0.9259259259259259
              precision    recall  f1-score   support

           0       0.89      0.67      0.76        12
           1       0.81      0.94      0.87        18

    accuracy                           0.83        30
   macro avg       0.85      0.81      0.82        30
weighted avg       0.84      0.83      0.83        30
```